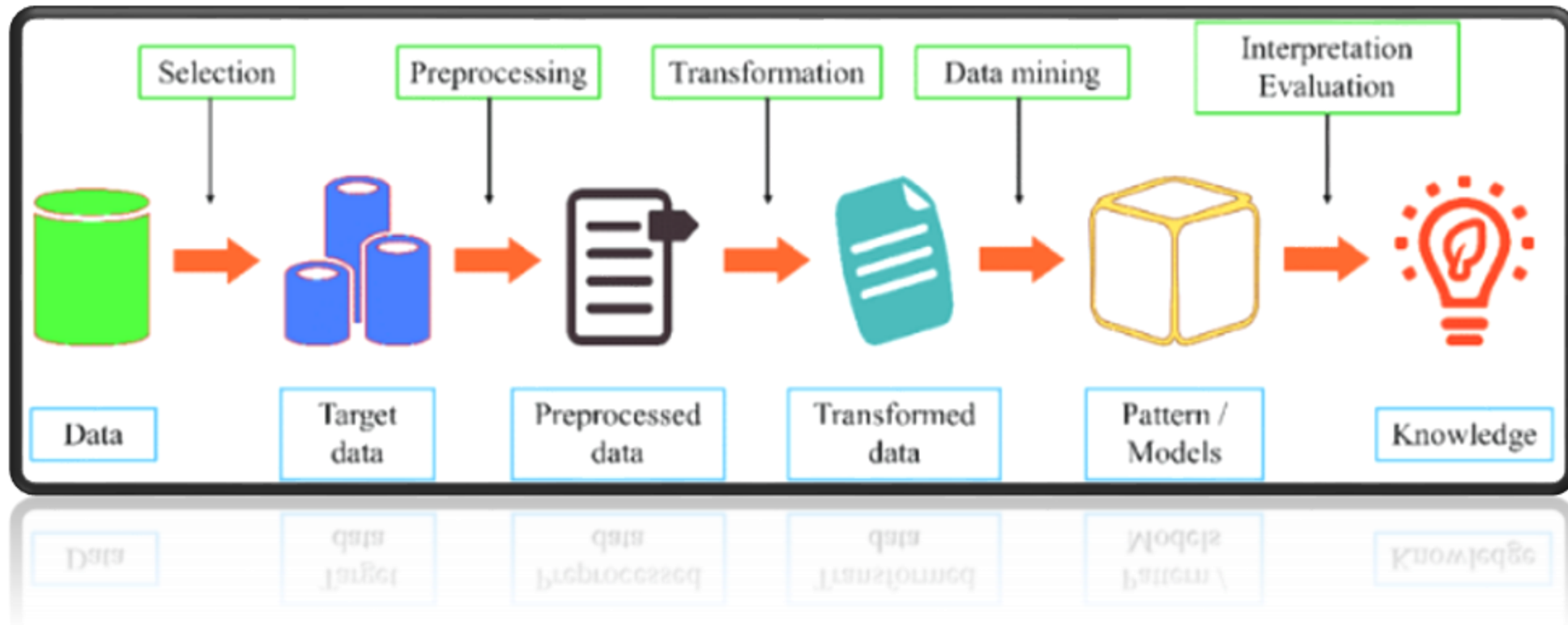


DATA MINING & DATA WAREHOUSE



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UNIT-1

Introduction to Data Mining: Why Data Mining?, What Is Data Mining?, What Kinds of Patterns Can Be Mined?, Which Technologies Are Used?, Which Kinds of Applications Are Targeted?, Major Issues in Data Mining.

Getting to Know your Data: Data Objects and Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity

Why Data Mining

- The Explosive Growth of Data: from terabytes to petabytes
 - **Data collection and data availability**
 - Automated data collection tools, database systems, and Web.
 - **Major sources of abundant data**
 - Business: Web, e-commerce, transactions, stocks, ...
 - Science: Remote sensing, bioinformatics, scientific simulation, ...
 - Society : news, digital cameras, YouTube

We are drowning in data, but starving for knowledge!

- “Necessity is the mother of invention”: Data mining ➡ Automated analysis of massive data sets

Evolution of Sciences

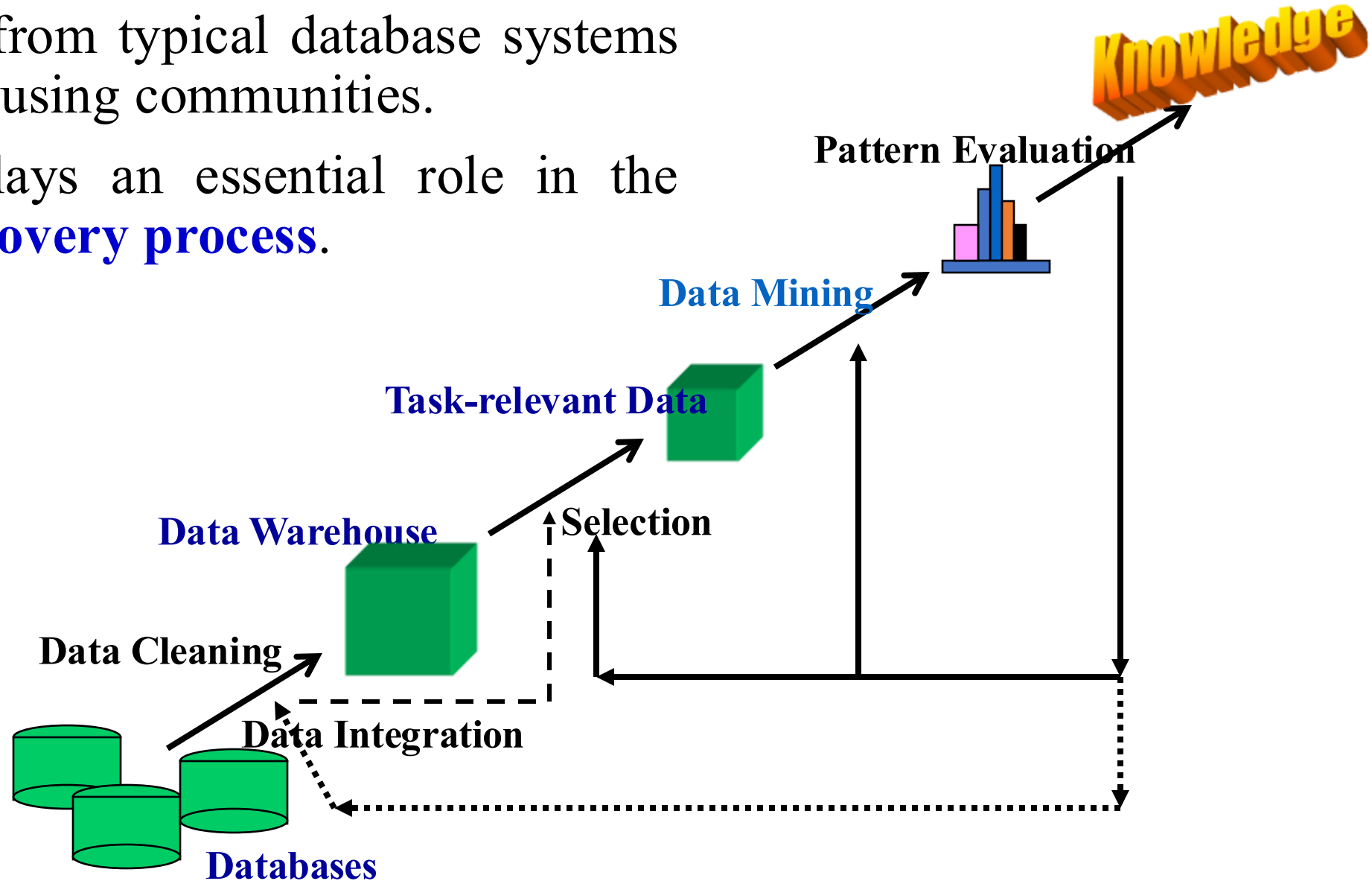
- Before 1600, **empirical science**
- 1600-1950s, **theoretical science**
 - Each discipline has grown a *theoretical component*. Theoretical models often motivate experiments and generalize our understanding.
- 1950s-1990s, **computational science**
 - Computational Science traditionally meant simulation. It grew out of our inability to find closed-form solutions for complex mathematical models.
- 1990-now, **data science**
 - The flood of data from new scientific instruments and simulations
 - The ability to economically store and manage petabytes of data online
 - **Data mining** is a major new challenge!

Evolution of Database Technology

- **1960s:**
 - Data collection, database creation, and network DBMS
- **1970s:**
 - Relational data model, relational DBMS implementation
- **1980s:**
 - RDBMS, advanced data models (extended-relational, OO, etc.)
- **1990s:**
 - Data mining, data warehousing, multimedia databases
- **2000s**
 - Data mining and its applications
 - **Web technology** (XML, data integration)

Knowledge Discovery (KDD) Process

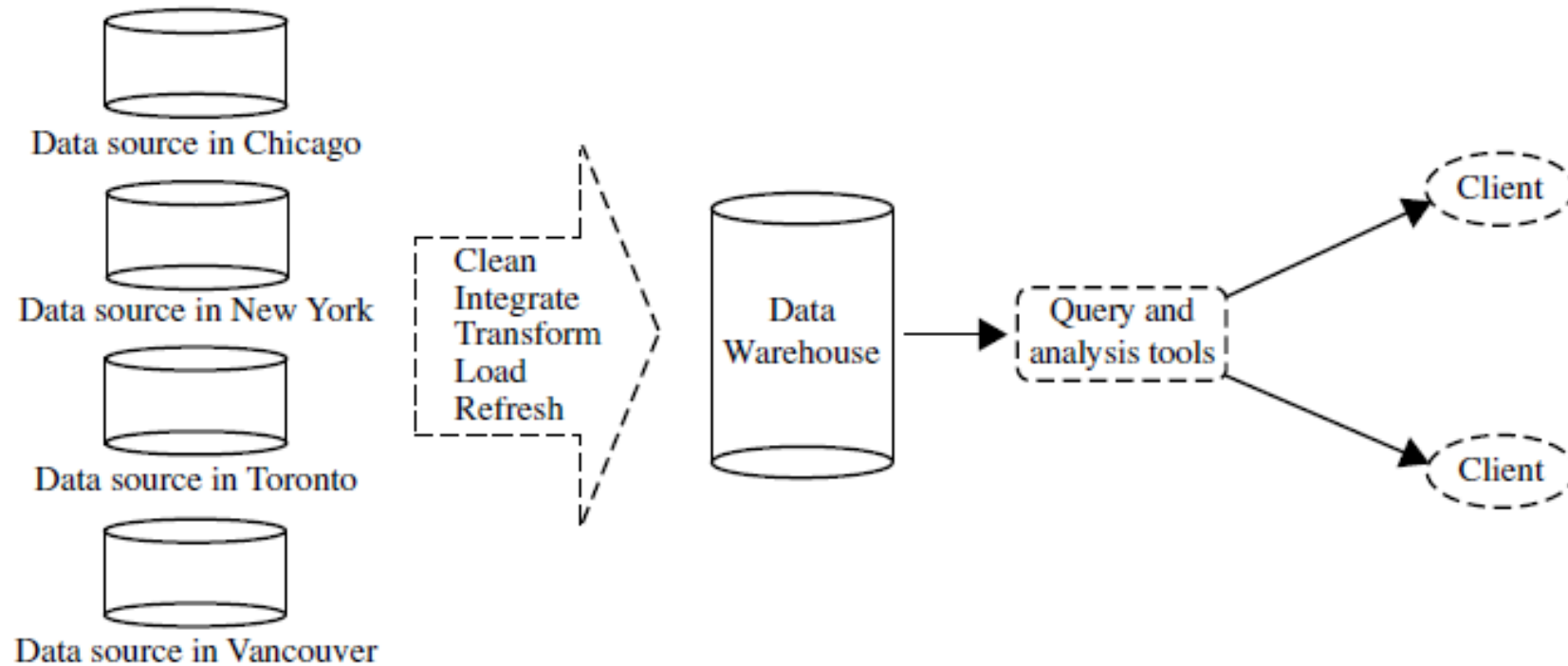
- This is a view from typical database systems and data warehousing communities.
- Data mining plays an essential role in the **knowledge discovery process**.



1. **Data cleaning** : To remove noise and inconsistent data
2. **Data integration** : Multiple data sources may be combined
3. **Data selection**: Data relevant to the analysis task are retrieved from the database.
4. **Data transformation**: Data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations.
5. **Data mining**: An essential process where intelligent methods are applied to extract data patterns.
6. **Pattern evaluation**: To identify the truly interesting patterns representing knowledge based on *interestingness measures*.
7. **Knowledge presentation**: Visualization and knowledge representation techniques are used to present mined knowledge to users.

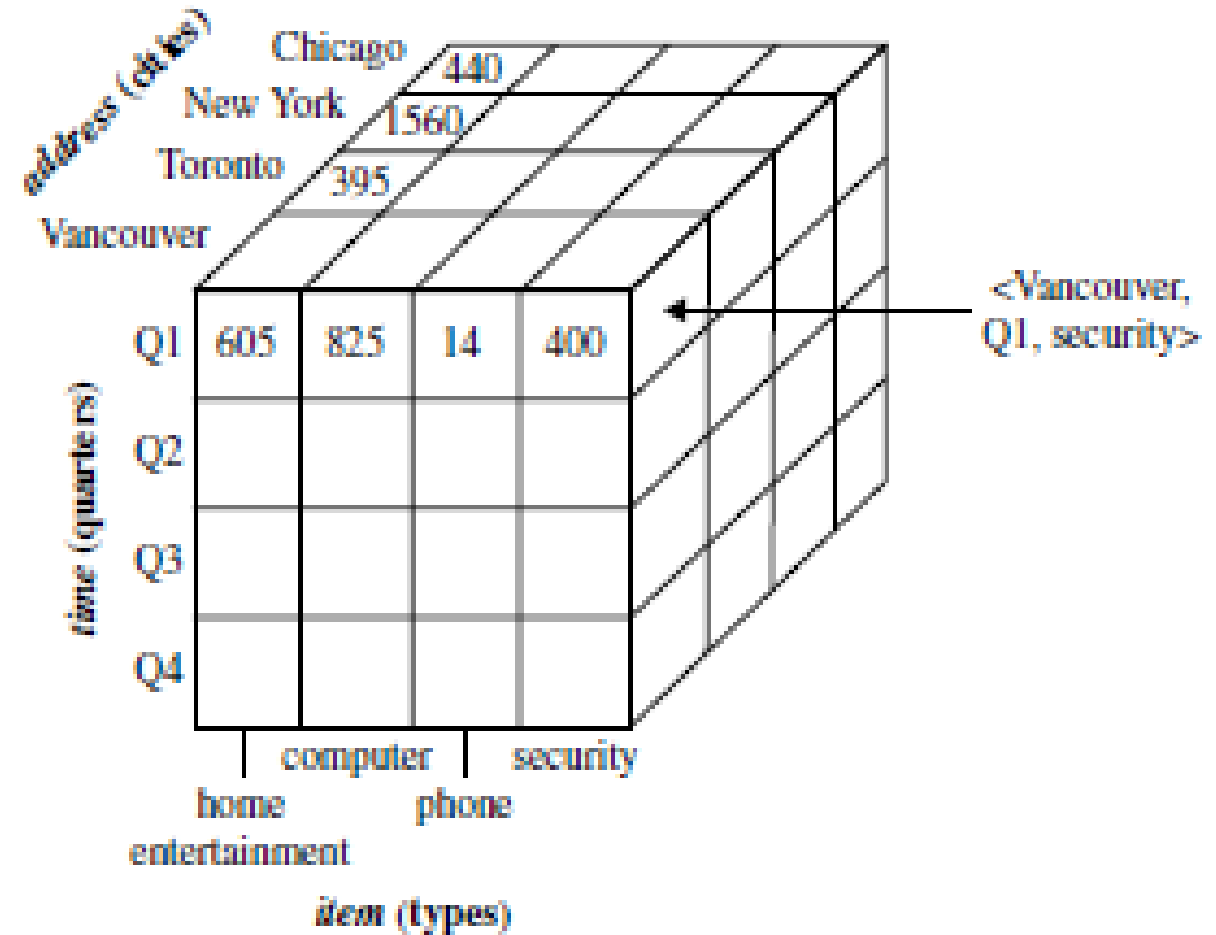
DATA WAREHOUSES

- A **data warehouse** is a repository of information **collected** from multiple sources, **stored** under a unified schema, and usually residing at a **single site**.
- Data warehouses are constructed via a process of data **cleaning**, data **integration**, data **transformation**, data **loading**, and periodic data **refreshing**.



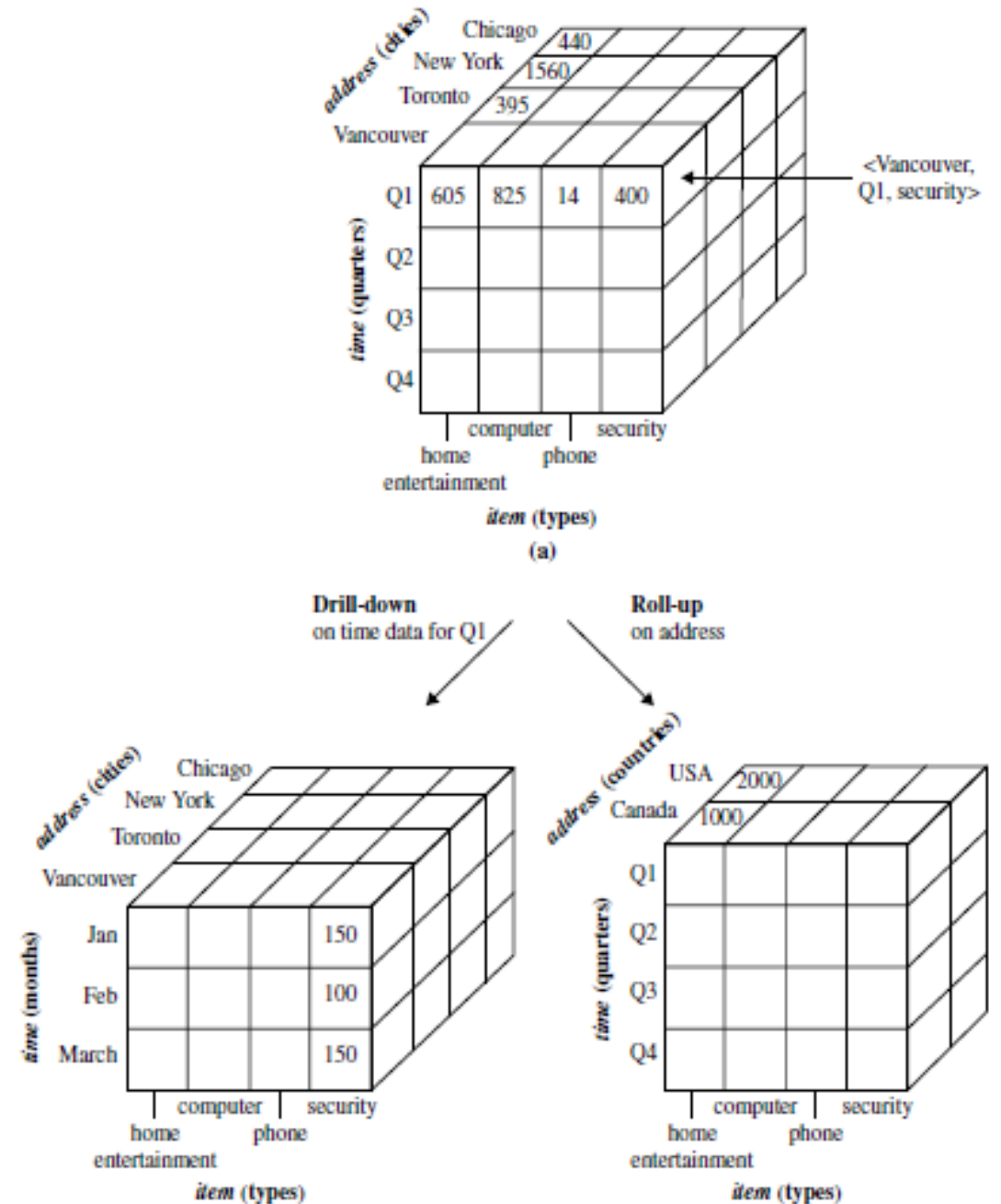
Typical framework of a data warehouse

- A **data warehouse** is usually modeled by a multidimensional data structure, called a **data cube**, in which each **dimension** corresponds to an attribute or a set of attributes in the schema, and each **cell** stores the value of some aggregate measure such as *count* or *sum(sales amount)*.
- A data cube provides a **multidimensional view of data** and allows the precomputation and fast access of summarized data.



Multidimensional Data Cube

- OnLine Analytical Processing operations (**OLAP**) allow the presentation of data at *different levels of abstraction*.
- Examples of OLAP operations: **drill-down** and **roll-up**.
- We can **drill down** on sales data summarized by *quarter* to see data summarized by *month*.
- We can **roll up** on sales data summarized by *city* to view data summarized by *country*.



Data Mining: On What Kinds of Data?

- **Database-oriented data sets**

- **R**elational database, **D**ata warehouse, **T**ransactional database

- **Advanced data sets**

- Data **S**treams and **S**ensor data
 - **T**ime-series data, **T**emporal data, sequence data (incl. bio-sequences)
 - **G**raphs, **S**ocial **N**etworks and multi-linked data
 - **S**patial data and **S**patiotemporal data
 - The **W**orld-Wide Web

What Kinds of Patterns Can Be Mined?

- There are a number of *data mining functionalities*:

Data characterization

- It is a summarization of the **general characteristics** or features of **a target class** of data.
- The output of data characterization can be presented in **various forms**.
- Examples: **pie charts, bar charts, curves, multidimensional data cubes**.

Data discrimination

- It is a **comparison** of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.
- Discrimination descriptions expressed in the form of rules are referred to as **discriminant rules**.

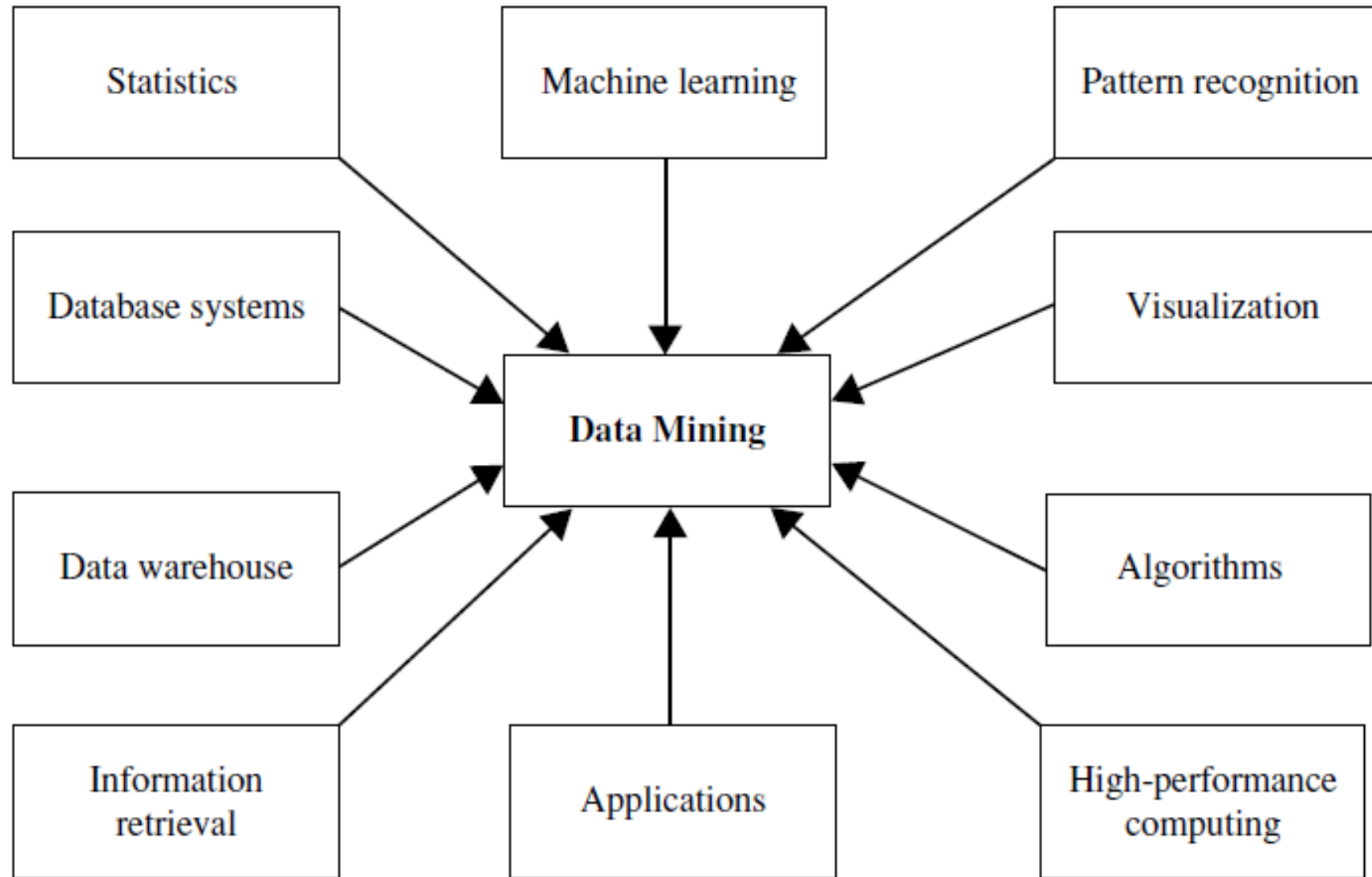
Mining Frequent Patterns, Associations, and Correlations

- **Frequent patterns**, as the name suggests, are patterns that occur frequently in data.
- A *frequent itemset* typically refers to a set of items that often appear together in a transactional data set (milk and bread).
- A frequently occurring subsequence, such as the pattern that customers, tend to purchase first a laptop, followed by a digital camera, and then a memory card, is a *(frequent) sequential pattern*.
- **Association analysis Rules:**

$buys(X, \text{"computer"}) \Rightarrow buys(X, \text{"software"})$ [support = 1%, confidence = 50%]

- A **confidence**, or certainty, of 50% means that if a customer buys a computer, **there is a 50% chance** that she will buy software as well.
- A 1% **support** means that 1% of all the transactions under analysis show that computer and software are **purchased together**.

Which Technologies Are Used



Applications of Data Mining

- **Web page analysis**: from web page classification, clustering to **PageRank & HITS algorithms**
- Collaborative analysis & **Recommender systems**
- **Basket data analysis** to targeted marketing
- **Biological and medical data analysis**: classification, cluster analysis (microarray data analysis), biological sequence analysis, biological network analysis
- Data mining and **software engineering** (e.g., IEEE Computer, Aug. 2009 issue)

Major Issues in Data Mining

- **Mining Methodology**

- Mining various and **new kinds** of knowledge
- Mining knowledge in **multi-dimensional space**
- Data mining: **An interdisciplinary effort**
- Boosting the power of discovery in a networked environment
- **Handling noise, uncertainty**, and incompleteness of data
- **Pattern evaluation** and pattern- or constraint-guided mining

- **User Interaction**

- Interactive mining
- Incorporation of background knowledge
- **Presentation and visualization** of data mining results

Major Issues in Data Mining

- **Efficiency and Scalability**

- **Efficiency** and **scalability** of data mining algorithms
- **Parallel, distributed**, and **incremental** mining methods

- **Diversity of data types**

- Handling **complex types of data**
- Mining dynamic, networked, and global data repositories

- **Data mining and society**

- **Social impacts** of data mining
- **Privacy-preserving** data mining
- **Invisible** data mining

Getting to Know your Data

Types of Data Sets

- **Record**

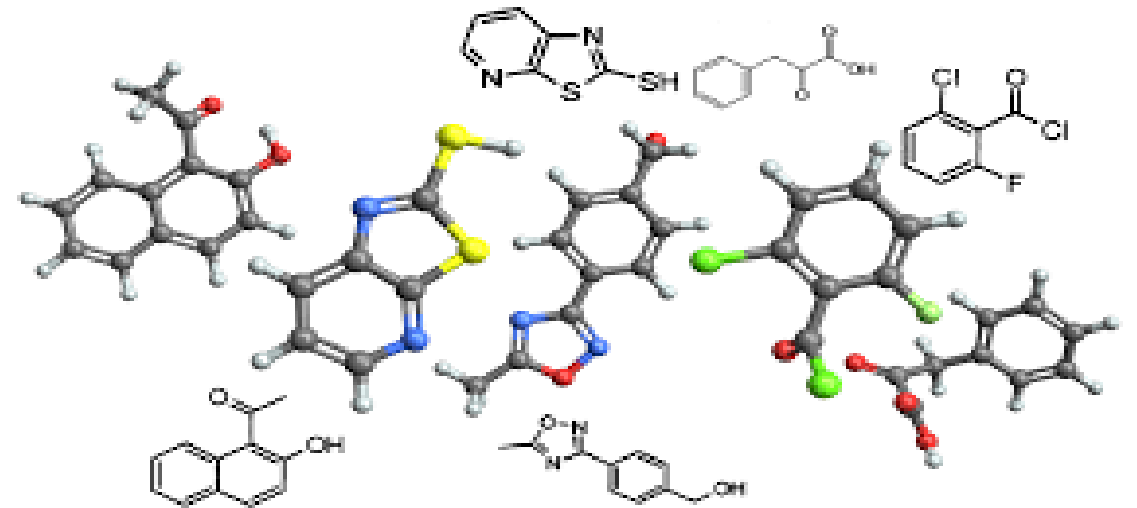
- Relational records
- Data matrix, e.g., numerical matrix, **crosstabs**
- Text documents
- Transaction data

- **Graph and network**

- World Wide Web
- Social or information networks
- Molecular Structures

Anxiety * Tension Crosstabulation

			Tension		Total
			low	high	
Anxiety	low	Count	5	21	26
		% within Anxiety	19.2%	80.8%	100.0%
		% within Tension	25.0%	75.0%	54.2%
		% of Total	10.4%	43.8%	54.2%
	high	Count	15	7	22
		% within Anxiety	68.2%	31.8%	100.0%
		% within Tension	75.0%	25.0%	45.8%
		% of Total	31.3%	14.6%	45.8%
Total	Count	20	28	48	
	% within Anxiety	41.7%	58.3%	100.0%	
	% within Tension	100.0%	100.0%	100.0%	
	% of Total	41.7%	58.3%	100.0%	



Data Objects

- **Data sets** are made up of **data objects**.
- A data object **represents** an **entity**.
- Examples:
 - **Sales database**: customers, store items, sales
 - **Medical database**: patients, treatments
 - **University database**: students, professors, courses
- Also called *samples , examples, instances, data points, objects, tuples*.
- Data objects are **described** by **attributes**.

Rows $\longrightarrow$ **Objects**; **Columns** $\longrightarrow$ **Attributes**.

Attributes

- **Attribute** (or **dimensions, features, variables**): **a data field**, representing a **characteristic** or **feature** of a data object.

E.g., customer_ID, name, address

- **Types:**
 - **Nominal** : *Gender, Marital status*
 - **Binary**: Boolean values----- *patient smoking*
 - **Numeric**: Quantitative
 - **Interval-scaled** : Measured on a scale of equal-size units
 - **Ratio-scaled** : A value is multiple(ratio) of another value.

Attribute Types

- **Nominal:** categories, states, or “names of things”
 - marital status, occupation
- **Binary**
 - **Nominal attribute with only 2 states** (0 and 1)
 - Symmetric binary: both outcomes **equally important**
e.g., *gender*
 - Asymmetric binary: outcomes **not equally important**.
e.g., *medical test (positive vs. negative)*
- **Ordinal**
 - Values have a **meaningful order** (ranking) but magnitude between successive values is not known.
e.g. *Size = {small, medium, large}*

Numeric Attribute Types

- **Quantity** (integer or real-valued)
- **Interval**
 - Measured on a scale of **equal-sized units**
E.g., *temperature in C° or F°*
 - **No true zero-point**
- **Ratio**
 - Inherent **zero-point**
 - We can speak of values as being an **order of magnitude larger than** the unit of measurement.
E.g., *length*

Discrete vs. Continuous Attributes

- **Discrete Attribute**

- Has only a **finite or countably infinite set** of value

E.g., zip codes

- **Continuous Attribute**

- Has **real numbers** as attribute values

E.g., temperature, height, or weight

- Continuous attributes are **typically represented** as **floating-point** variables

BASIC STATISTICAL DESCRIPTIONS OF DATA

- Motivation

- To **better understand the data**
- Central tendency, variation and spread

- Data dispersion characteristics

- median, max, min, quantiles.

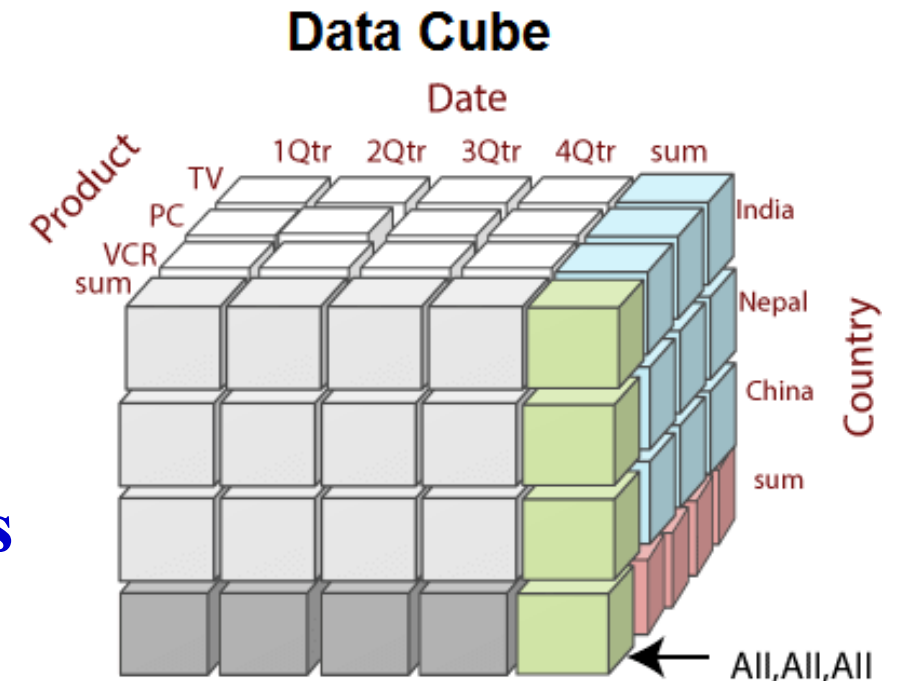
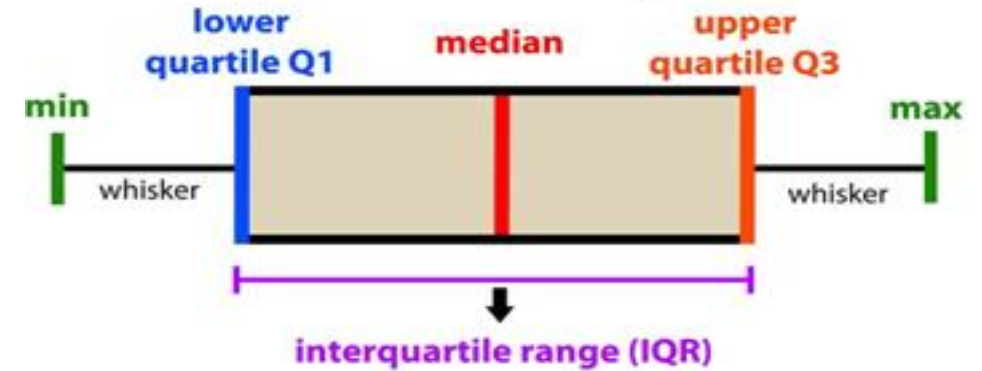
- Numerical dimensions (sorted intervals)

- **Data dispersion**: The extent to which a distribution is **stretched** or **squeezed**.

- **Boxplot** or **quantile analysis**

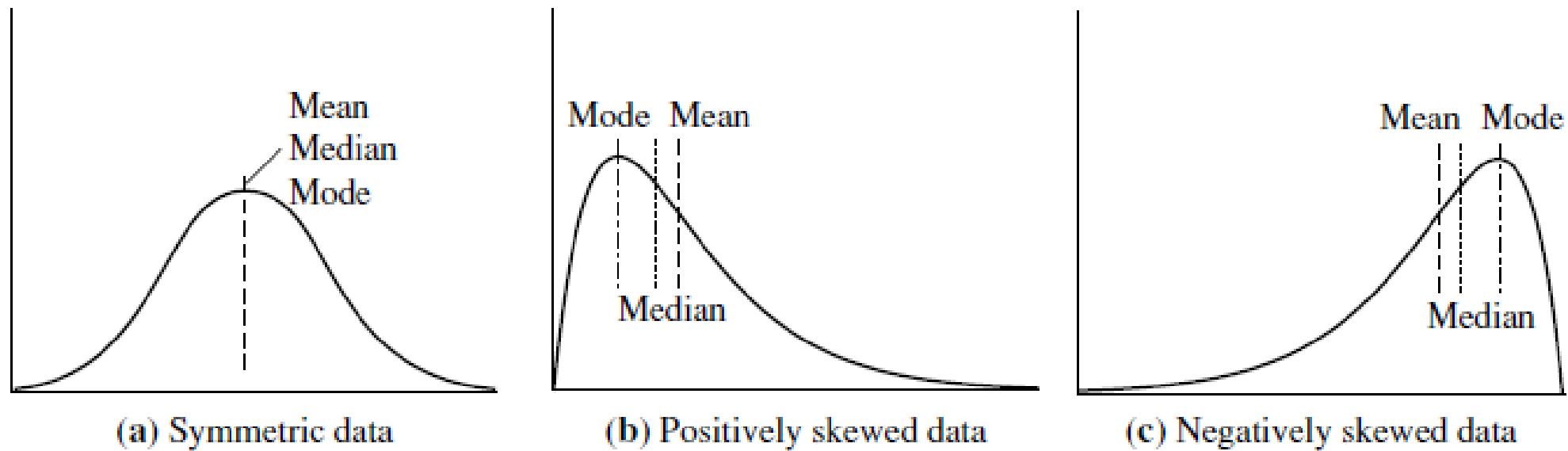
- Dispersion analysis on computed measures

- Folding statics into **numerical dimensions**
- **Transformed cube** on values



Symmetric vs. Skewed Data

- Median, Mean and Mode of symmetric, positively and negatively skewed data.



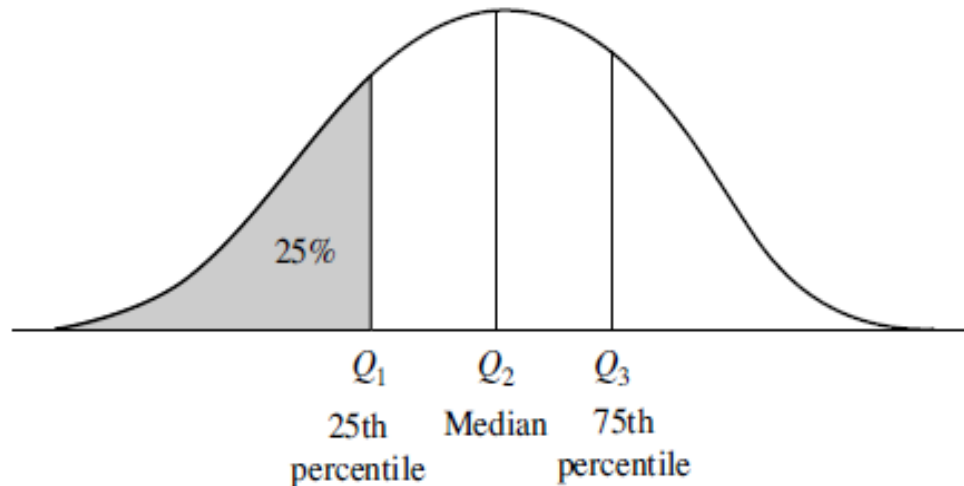
Mean, median, and mode of symmetric versus positively and negatively skewed data.

Measuring the Dispersion of Data

- To assess the **dispersion** or **spread** of numeric data.
- **Measures:** **R**ange, **Q**uantiles, **Q**uartiles, and **P**ercentiles.
- **Range:** The **difference between** the **largest** (max()) and **smallest** (min()) values.
- **Quantiles:** **Points taken at regular intervals** of a data distribution, dividing it into essentially equalize consecutive sets.

E.g: 2-Quantile —————→ *Median*

E.g: 4-Quantile —————→ **Three** data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution. They are more commonly referred to as **Quartiles**.



First Quartile: Q_1 : 25th percentile

Second Quartile: Q_2 : Median

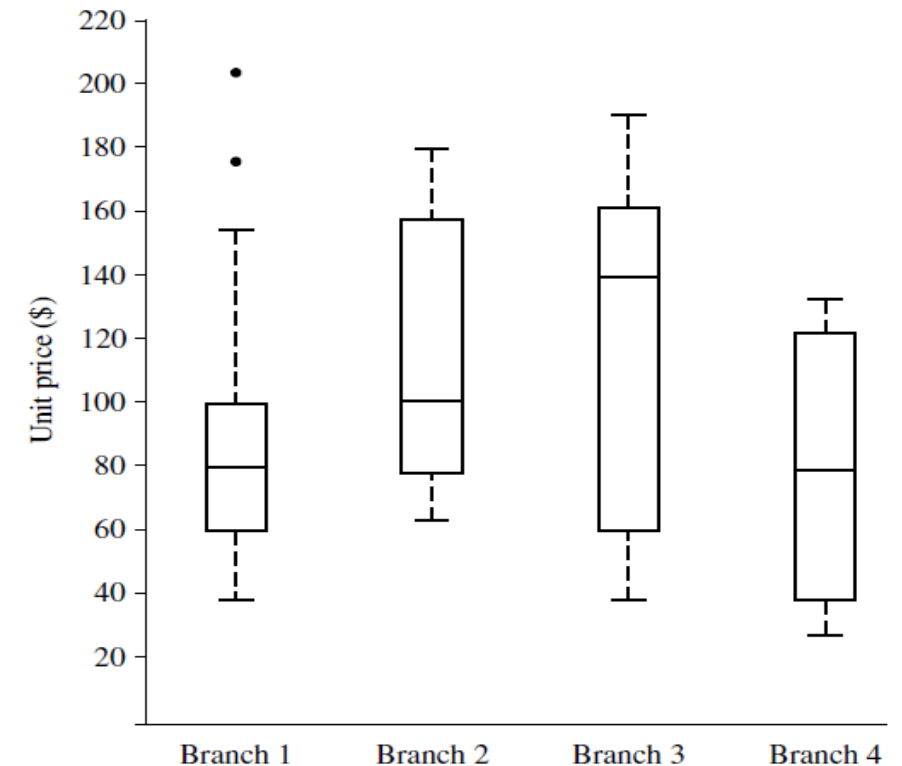
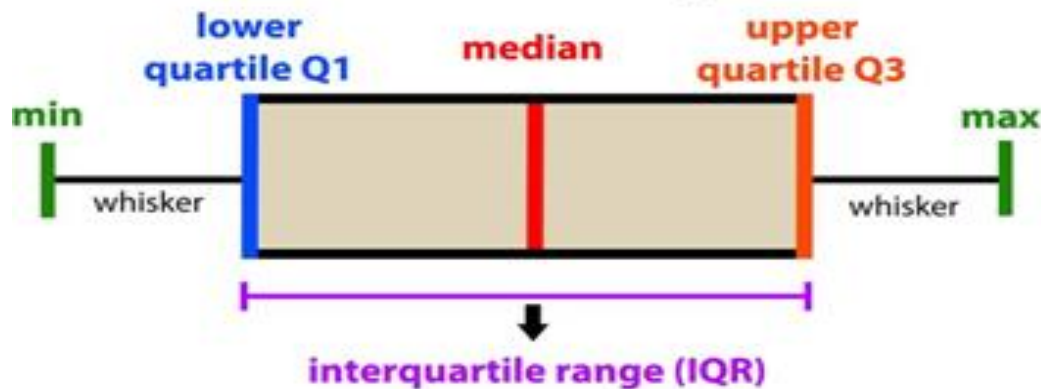
Third Quartile: Q_3 : 75th percentile

- **Inter Quartile Range (IQR):** The **distance** between the first and third quartiles.

$$\text{IQR} = Q_3 - Q_1$$
 - **Five-Number Summary:** No single numeric measure of spread is very useful.
- A fuller summary of the shape of a distribution:


Minimum, **Q1**, **Q2(Median)**, **Q3**, **Maximum**


- **Boxplots:** To Visualizing a distribution.
 - **Two ends** of the box are at the quartiles.
 - The **median** is marked by a line within the box
 - **Two lines** (*whiskers*) outside the box extend to the smallest and largest observations

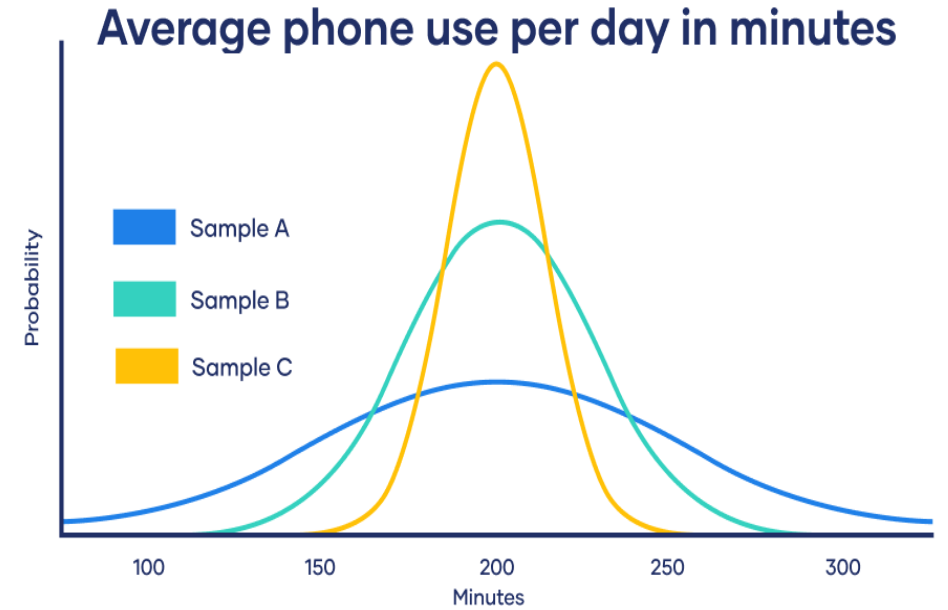


Variance and Standard Deviation

- Measures of **data dispersion**.

SD(**Low**): Data observation is close to **Mean**

SD(**High**): Data spread over a **large range**



The **variance** of N observations, $x_1, x_2, \dots, x_N$, for a numeric attribute X is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2,$$

- SD** is the square root of **Variance**.

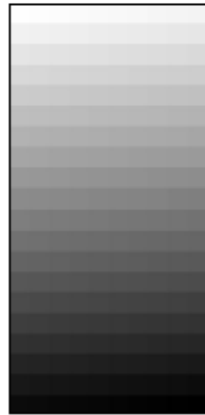
DATA VISUALIZATION

- To communicate data **clearly** and **effectively** through **Graphical Means**.
- **Pixel-Oriented Visualization Techniques** : **Colour of the pixel** reflects the dimensional value.

E.g. Four dimensions: income, credit limit, transaction volume, age.



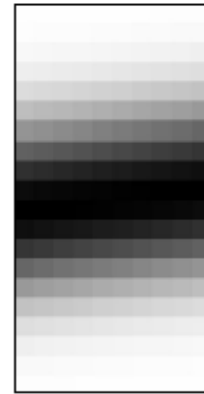
Sort all customers in income-ascending order



(a) *income*



(b) *credit_limit*



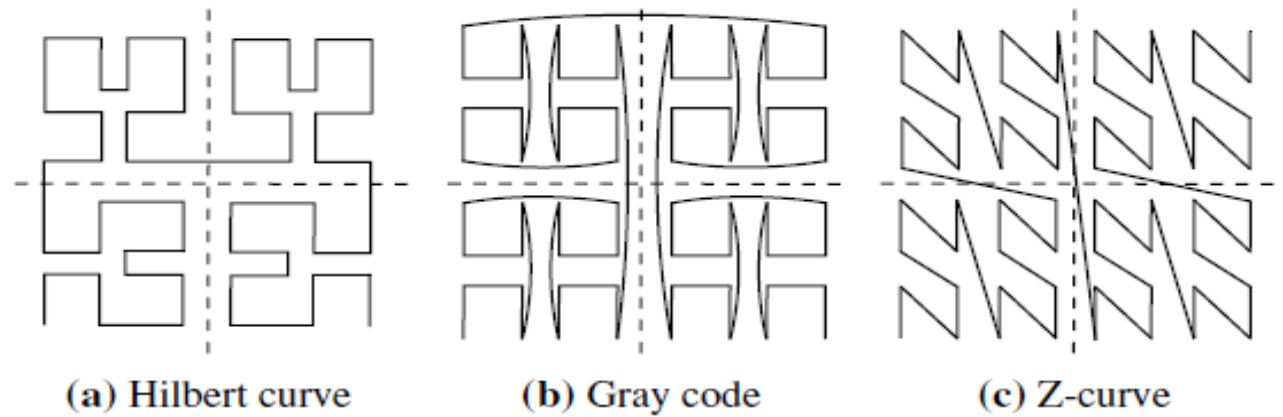
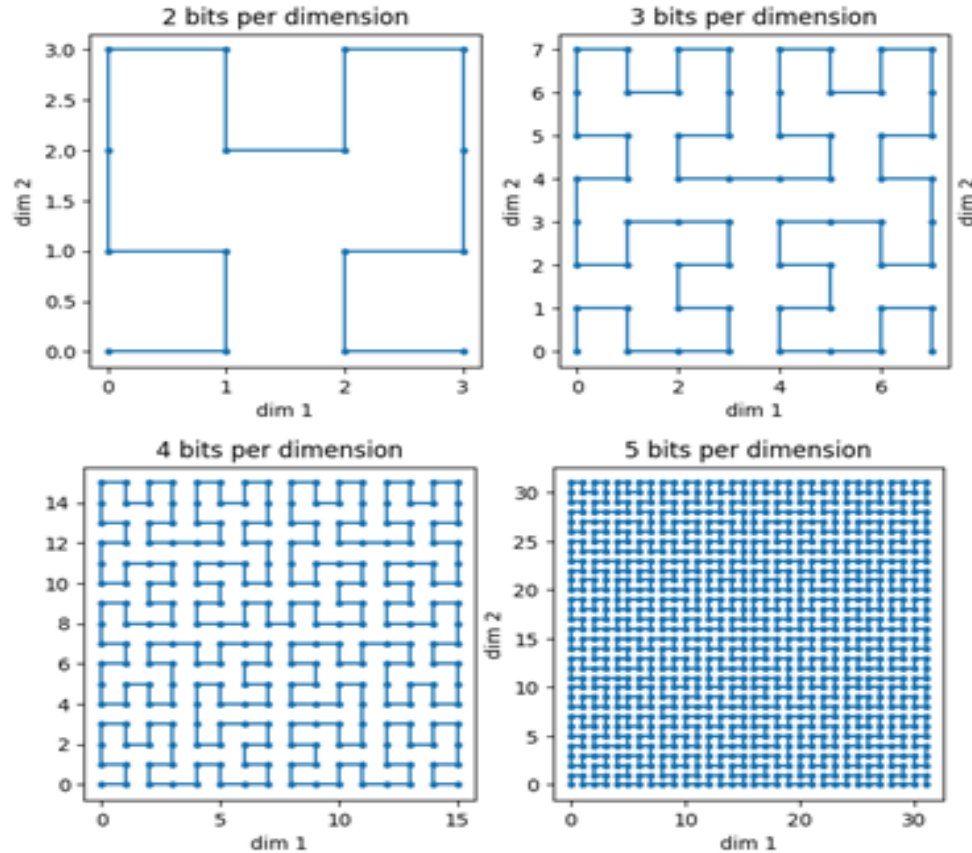
(c) *transaction_volume*



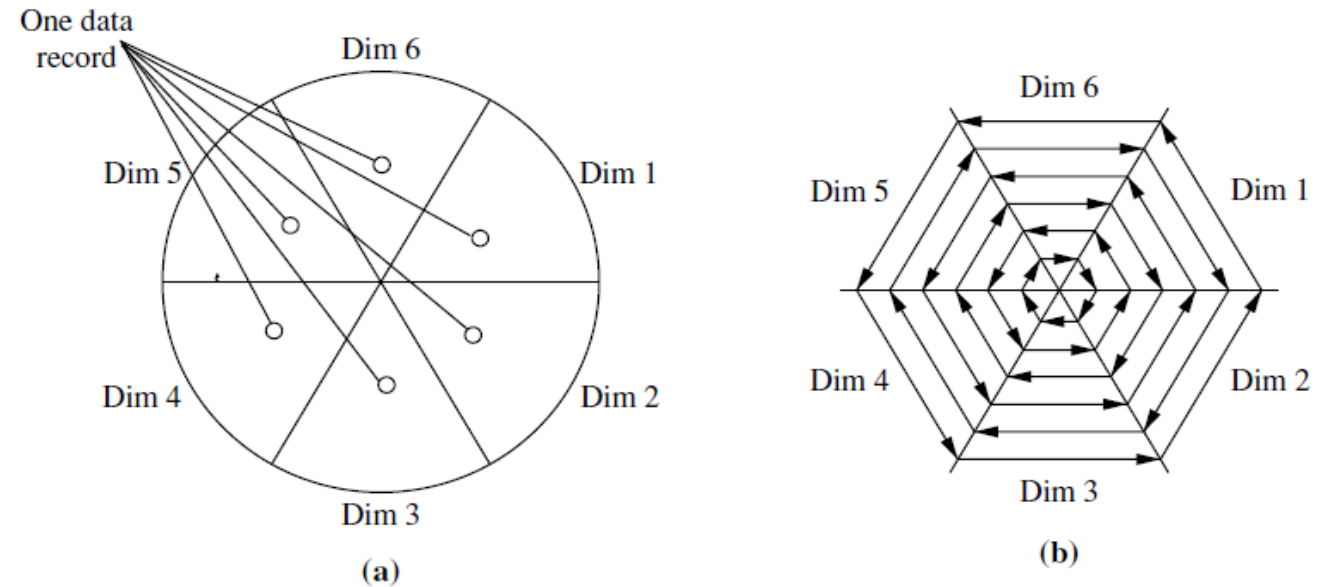
(d) *age*

- **Limitation:** A pixel is next to the one above it in the window, even though the two are not next to each other in the global order.

- A *space-filling curve* is a curve with a range that **covers the entire n-dimensional unit** hypercube.



Some frequently used 2-D space-filling curves.

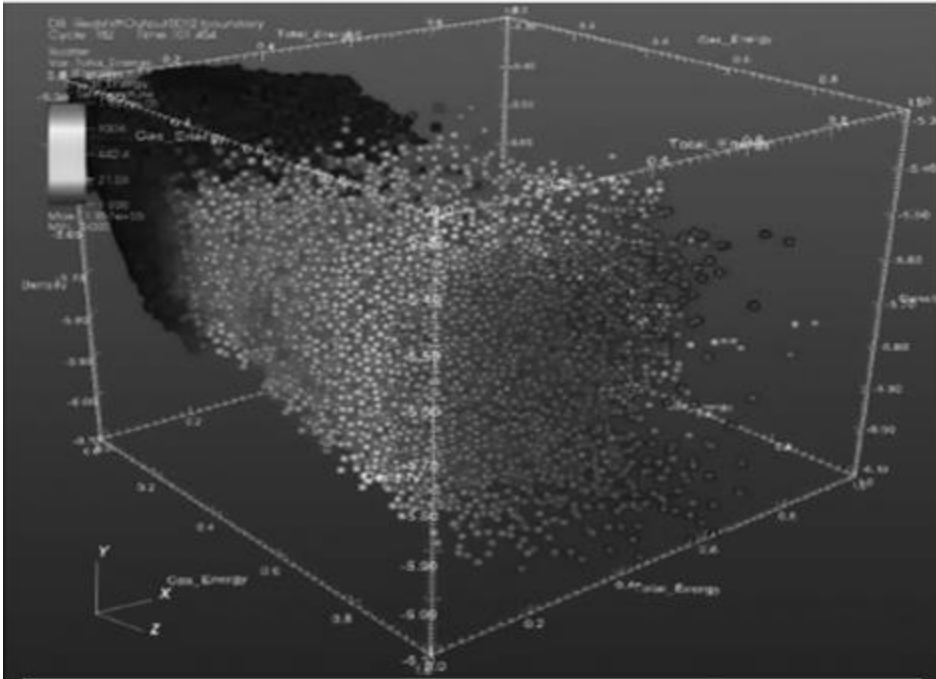


- The *circle segment technique* uses windows in the shape of **segments of a circle**

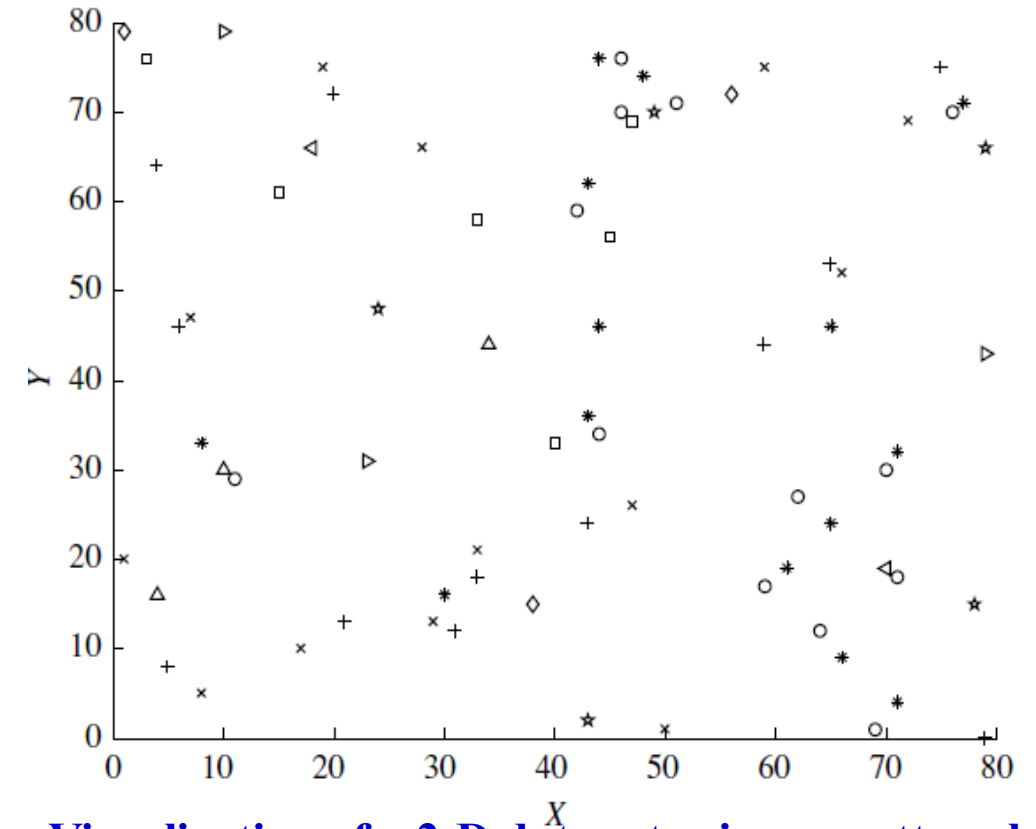
The circle segment technique. (a) Representing a data record in circle segments. (b) Laying out pixels in circle segments.

Geometric Projection Visualization Techniques

- To find **interesting projections** of multidimensional data sets
- A **scatter plot** displays **2-D** data points using Cartesian coordinates.
- **X** and **Y** are two spatial attributes and the third dimension is represented by **different shapes**.
- A **3-D** scatter plot uses three axes.

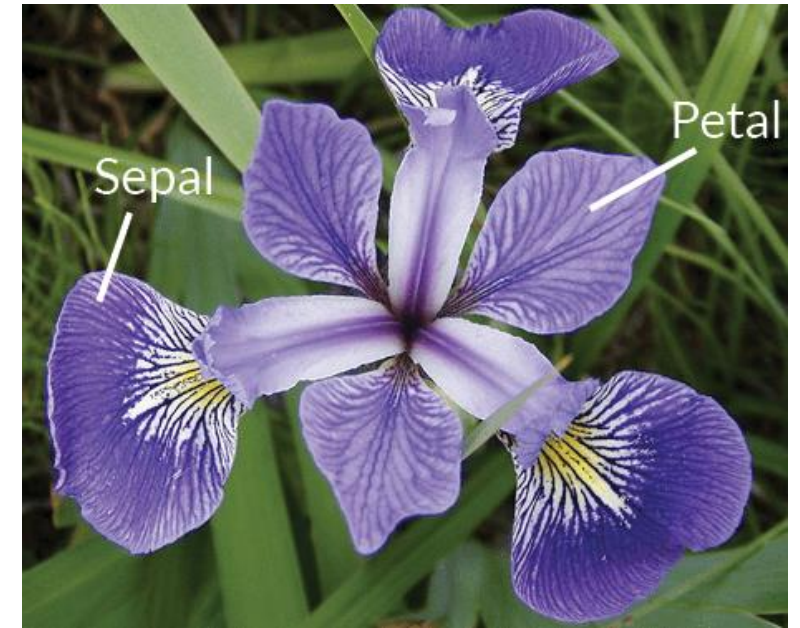
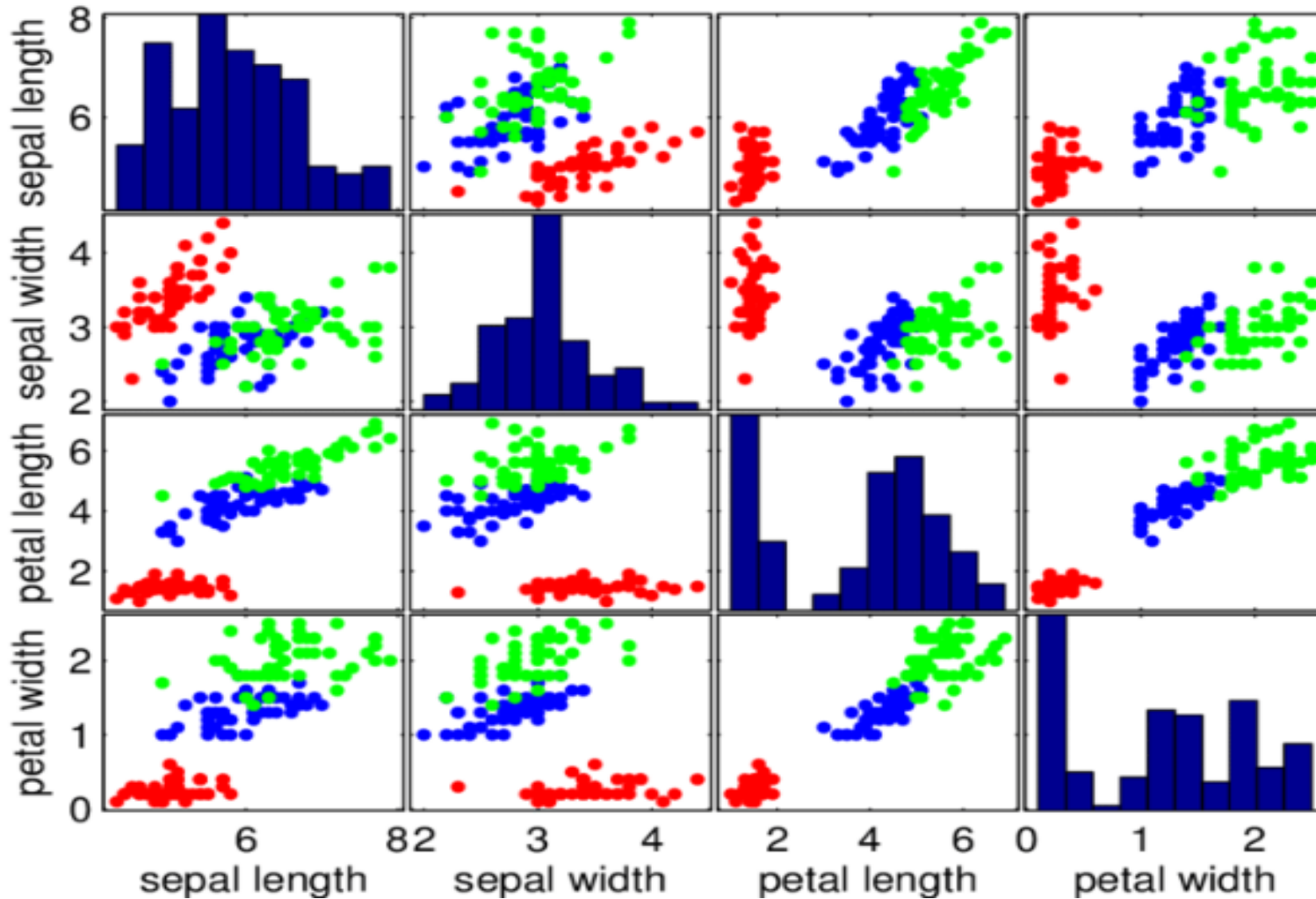


Visualization of a 3-D data set using a scatter plot



Visualization of a 2-D data set using a scatter plot

- The **scatter-plot matrix** technique is a useful **extension to the scatter plot**.



Iris Flower Sample

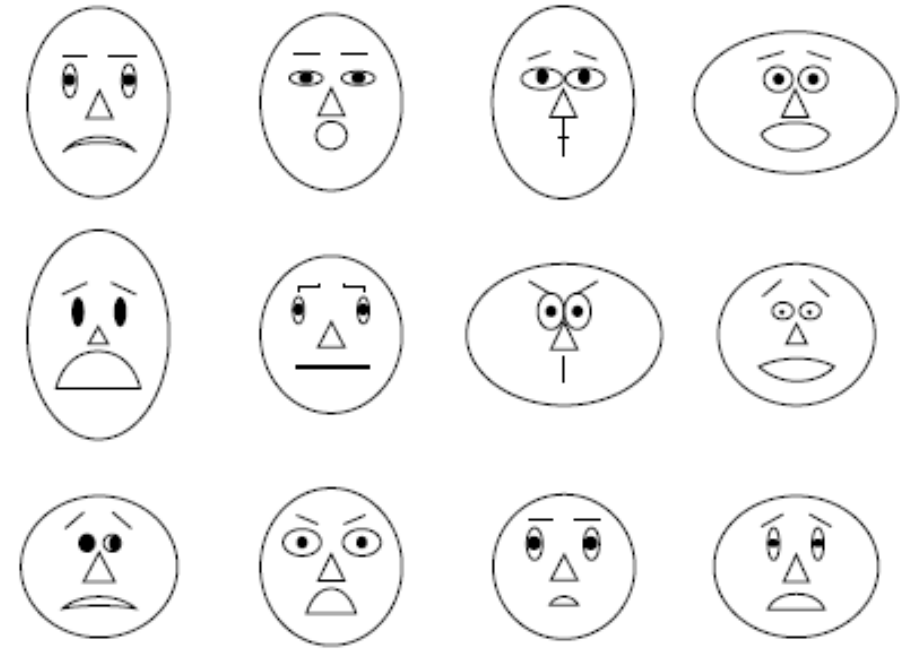
<http://archive.ics.uci.edu/dataset/53/iris>

Icon-Based Visualization Techniques:

- Use **small icons** to represent multidimensional data values.

1) Chernoff faces:

- Helps to reveal trends in the data using **various expressions**.



2) Stick figures:

- It maps multidimensional data to **five-piece stick figures**, where each figure has four limbs and a body.

Visualizing Complex Data and Relations

- To visualize **numerical** and **non-numerical** data.

❖ Tag cloud:

- A visualization of **statistics** of **user-generated tags**.
- The **importance** of a tag is indicated by **font size** or **color**.

animals architecture art asia australia autumn baby band barcelona beach berlin bike bird
birds birthday black blackandwhite blue bw california canada canon car cat
chicago china christmas church city clouds color concert cute dance day de dog
england europe fall family fashion festival film florida flower flowers food
football france friends fun garden geotagged germany girl girls graffiti green
halloween hawaii holiday home house india iphone ireland island italia italy japan july kids la
lake landscape light live london love macro me mexico model mountain mountains museum
music nature new newyork newyorkcity night nikon nyc ocean old paris
park party people photo photography photos portrait red river rock san
sanfrancisco scotland sea seattle show sky snow spain spring street summer
sun sunset taiwan texas thailand tokyo toronto tour travel tree trees trip uk urban
usa vacation washington water wedding white winter yellow york zoo

MEASURING DATA SIMILARITY AND DISSIMILARITY

- A **cluster** is a collection of data objects such that the objects within a cluster are similar to one another and dissimilar to the objects in other clusters.
- Knowledge of object **similarities** can also be used in nearest-neighbor classification schemes.
- Given object (e.g., a patient) is assigned a **class label** (diagnosis) based on its similarity toward other objects.

Data Matrix versus Dissimilarity Matrix

*Case : **n objects** (e.g., persons, items, or courses) described by **p attributes** (**features**) such as age, height, weight, or gender).*

- **Data matrix:**

object-by-attribute structure

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix :

object-by-object structure

- $d(i, j)$: dissimilarity or “difference” between objects i and j .

$$\begin{bmatrix} 0 & & & & \\ d(2, 1) & 0 & & & \\ d(3, 1) & d(3, 2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n, 1) & d(n, 2) & \dots & \dots & 0 \end{bmatrix}$$

- **Similarity** can often be expressed as a function of measures of **Dissimilarity**:

$$\text{sim}(i, j) = 1 - d(i, j),$$

Proximity Measures for Nominal Attributes

- A nominal attribute can take on two or more states.
- E.g.. **map color** is a nominal attribute (five states): **red**, **yellow**, **green**, **pink**, and **blue**
States can be denoted by letters, symbols, or a set of integers, such as **1**, **2**, **:::**, **M**.

$$d(i, j) = \frac{p - m}{p}$$

m is the number of matches

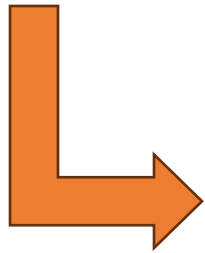
p is the total number of attributes

- Example: Table with **mixed type data**

Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

$$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ d(3,1) & d(3,2) & 0 & \\ d(4,1) & d(4,2) & d(4,3) & 0 \end{bmatrix}$$

$$d(i,j) = \frac{p-m}{p}$$



$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

- Conclusion:** All objects are dissimilar except objects 1 and 4 (i.e., $d(1,4)=0$).

Proximity Measures for Binary Attributes:

- Binary attribute has **Two** states: **0** and **1** $\longrightarrow$ **A**bsent and **P**resent.
- To calculate dissimilarity between binary attributes: **Contingency Table**

q : No. of **attributes** that equal to 1
for both objects i and j.

r : No. of attributes that equal to 1
for object i and 0 for object j.

Similarly, **s** and **t** value.

p : Total no. of attributes = **q+r+s+t**

❖ Distance measure for **symmetric binary variable**

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

❖ Distance measure for **asymmetric binary variables**:

$$d(i, j) = \frac{r + s}{q + r + s}$$

❖ Jaccard coefficient (**similarity** measure for
asymmetric binary variables):

$$sim_{Jaccard}(i, j) = \frac{q}{q + r + s}$$

Contingency Table for Binary Attributes

	Object j			sum
	1	0		
Object i	1	q	r	$q + r$
	0	s	t	$s + t$
	sum	$q + s$	$r + t$	p

Dissimilarity of Numeric Data: Minkowski Distance

- The dissimilarity of objects described by numeric attributes.

1) Euclidean distance:

Let $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ be two objects described by p numeric attributes. The Euclidean distance between objects i and j is defined as

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ip} - x_{jp})^2}$$

2) Manhattan distance : It can be defined as,

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

❖ The **properties** of Euclidean and Manhattan:

Non-negativity: $d(i, j) \geq 0$: Distance is a non-negative number

Identity of indiscernible: $d(i, i) = 0$: Distance of object to itself is 0.

Symmetry: $d(i, j) = d(j, i)$: Distance is symmetric function.

Triangle inequality: $d(i, j) \leq d(i, k) + d(k, j)$: Going directly from object i to object j in space is no more than making a detour over any other object k .

Cosine Similarity

- A **document** can be represented by **thousands of attributes**, each recording the frequency of a particular word.

Document Vector or Term-Frequency Vector

Document	team	coach	hockey	baseball	soccer	penalty	score	win	loss	season
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Cosine similarity** is a measure of similarity that can be used to compare documents.

Let **x** and **y** be **two vectors**. Then **similarity** will be : $\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$

$\|\mathbf{x}\|$: Euclidian norm of **x** vector = $\sqrt{x_1^2 + x_2^2 + x_3^2 + \dots + x_p^2}$

E.g. $\mathbf{x} = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$ and $\mathbf{y} = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$

How similar are $\mathbf{x}$ and $\mathbf{y}$?

$$\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

$$\begin{aligned} \mathbf{x}^t \cdot \mathbf{y} &= 5 \times 3 + 0 \times 0 + 3 \times 2 + 0 \times 0 + 2 \times 1 + 0 \times 1 + 0 \times 0 + 2 \times 1 \\ &\quad + 0 \times 0 + 0 \times 1 = 25 \end{aligned}$$

$$\|\mathbf{x}\| = \sqrt{5^2 + 0^2 + 3^2 + 0^2 + 2^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2} = 6.48$$

$$\|\mathbf{y}\| = \sqrt{3^2 + 0^2 + 2^2 + 0^2 + 1^2 + 1^2 + 0^2 + 1^2 + 0^2 + 1^2} = 4.12$$

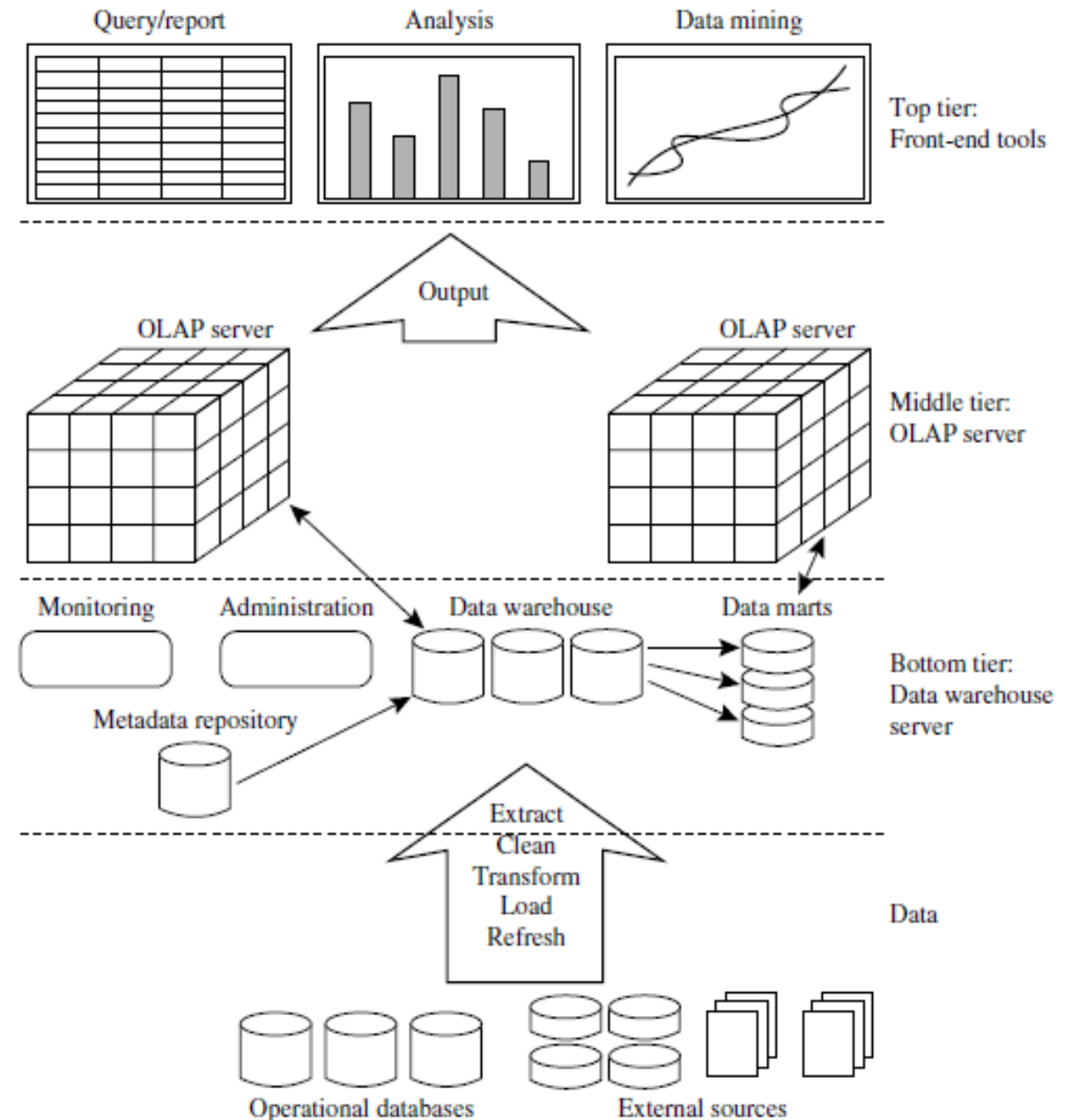
$$\text{sim}(\mathbf{x}, \mathbf{y}) = 0.94$$

Document 1  **Document 2** are considered to be *quite similar*.

Data Warehousing: A Multitiered Architecture

Data warehouses often adopt a **three-tier architecture**:

- The **bottom tier** is a **warehouse database server** that is a relational database system.
- The **middle tier** is an **OLAP server** that is typically implemented using either:
 - (1) A **relational OLAP (ROLAP)** model .
 - (2) A **multidimensional OLAP (MOLAP)** model.
- The **top tier** is a **front-end client layer**, which contains query and reporting tools, analysis, and data mining tools.



Data Warehouse Models: Enterprise Warehouse, Data Mart, and Virtual Warehouse

1. Enterprise warehouse:

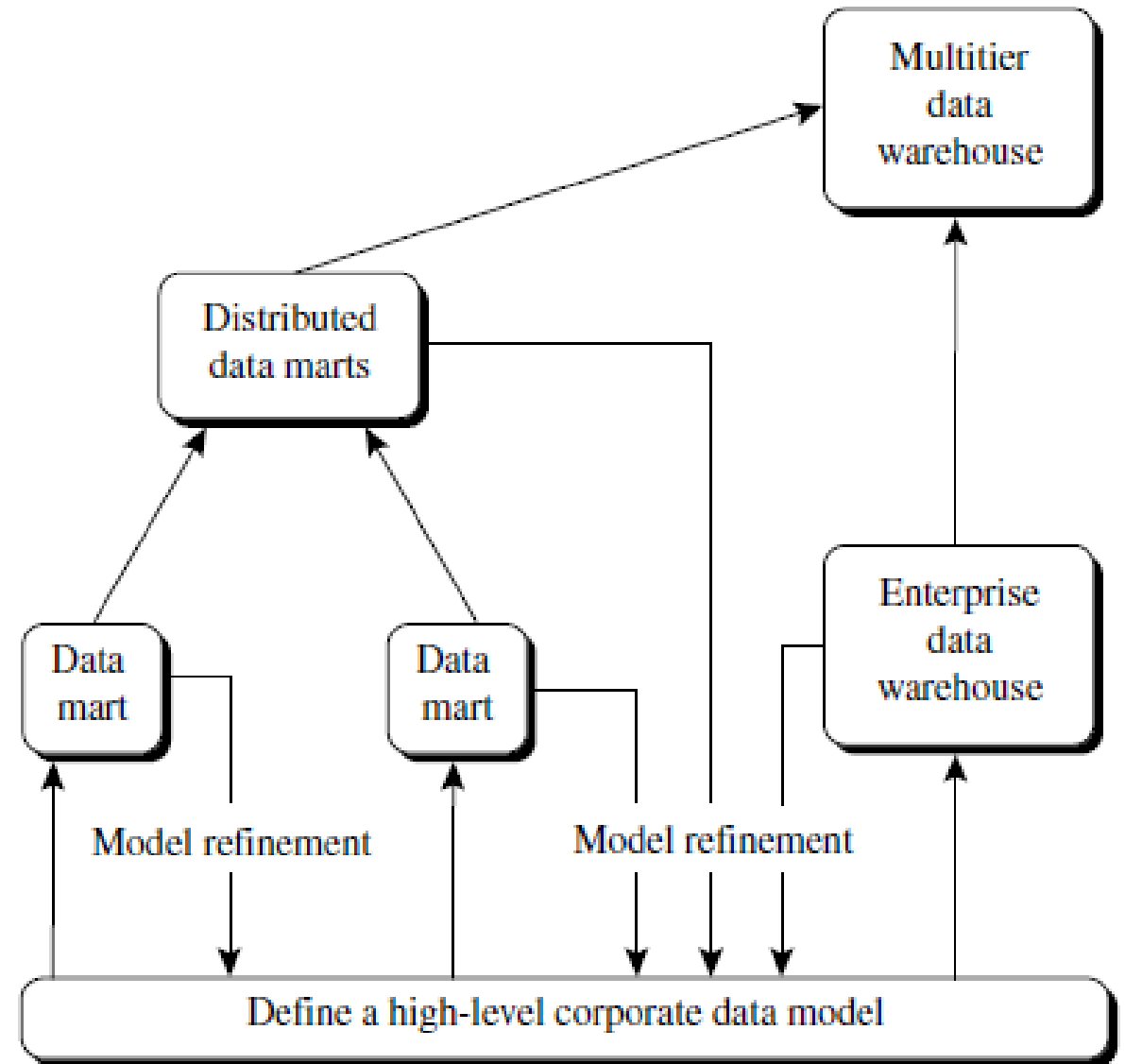
- It collects all of the information about subjects **spanning the entire organization.**
- It provides **corporate-wide data integration.**
- It typically contains **detailed data** as well as **summarized data**
- An enterprise data warehouse may be implemented on **traditional mainframes.**

2. Data mart:

- A data mart contains **a subset of corporate-wide data** of a specific group of users.
- The **scope** is confined to **specific selected subjects**. Example: a marketing data .
- The data contained in data marts tend **to be summarized.**
- **Independent data marts** are sourced from data captured from one or more operational systems.
- **Dependent data marts** are sourced directly from enterprise data warehouses.

3. Virtual warehouse:

- It is a **set of views** over operational databases for efficient query processing,
- **Easy to build** but requires **excess capacity** on operational database servers.



! A recommended approach for data warehouse development.

Extraction, Transformation, and Loading

- Data warehouse systems use **back-end tools** and **utilities** to populate and refresh their data:
 - **Data extraction**, which typically **gathers data** from multiple, heterogeneous, and external sources.
 - **Data cleaning**, which **detects errors** in the data and rectifies them when possible.
 - **Data transformation**, which **converts data** from legacy or host format to warehouse format.
 - **Load**, which **sorts, summarizes, consolidates**, computes views, checks integrity, and builds indices and partitions.
 - **Refresh**, which **propagates the updates** from the data sources to the warehouse.

Metadata Repository

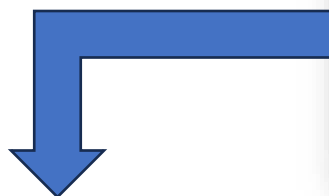
- **Metadata** are **data about data**. It contain the following:
 1. A **description** of the **data warehouse structure**.
 2. **Operational metadata**, which include **currency** of data.
 3. The **algorithms used for summarization**, which include **measure** and **dimension** definition algorithms.
 4. **Mapping from the operational environment to the data warehouse**, which **includes source databases** and **their contents**
 5. **Data related to system performance**, which include **indices** and **profiles that improve data** access and retrieval
 6. **Business metadata**, which include business terms and definitions, data **ownership information**, and **charging policies**.

Data Warehouse Modeling: Data Cube and OLAP

- Data warehouses and OLAP tools are based on a **multidimensional data model**.
- A **Data Cube** allows data to be modeled and **viewed** in **multiple dimensions**.
- **Dimensions** are the perspectives or entities with respect to which an organization wants to keep records. **E.g. Sales data.**
- Each dimension may have a table associated with it, called a **Dimension Table**, which **further describes the dimension**. E.g. Item  name, brand, type
- **Fact Table:** Theme of the multidimensional data. E.g.: **Sales**
- Data warehousing the **data cube** is ***n*-dimensional**.
- The **3-D data** in the table are represented as a series of **2-D tables**.
- We can think of a **4-D cube** as being a series of **3-D cubes**.
- We may display any ***n*-dimensional** data as a series of ***n*-1 dimensional** “cubes.”

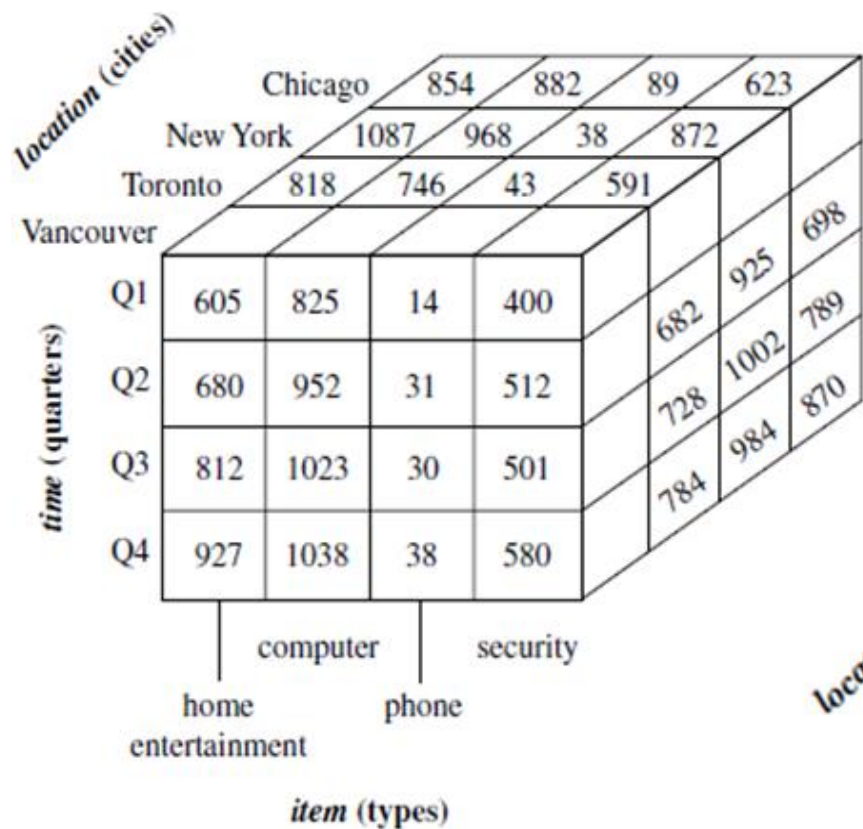
2-D View of Sales Data for *AllElectronics* According to *time* and *item*

<i>location</i> = "Vancouver"				
<i>time</i> (quarter)	<i>item</i> (type)			
	<i>home entertainment</i>	<i>computer</i>	<i>phone</i>	<i>security</i>
Q1	605	825	14	400
Q2	680	952	31	512
Q3	812	1023	30	501
Q4	927	1038	38	580

3-D View of Sales Data for *AllElectronics* According to *time*, *item*, and *location*

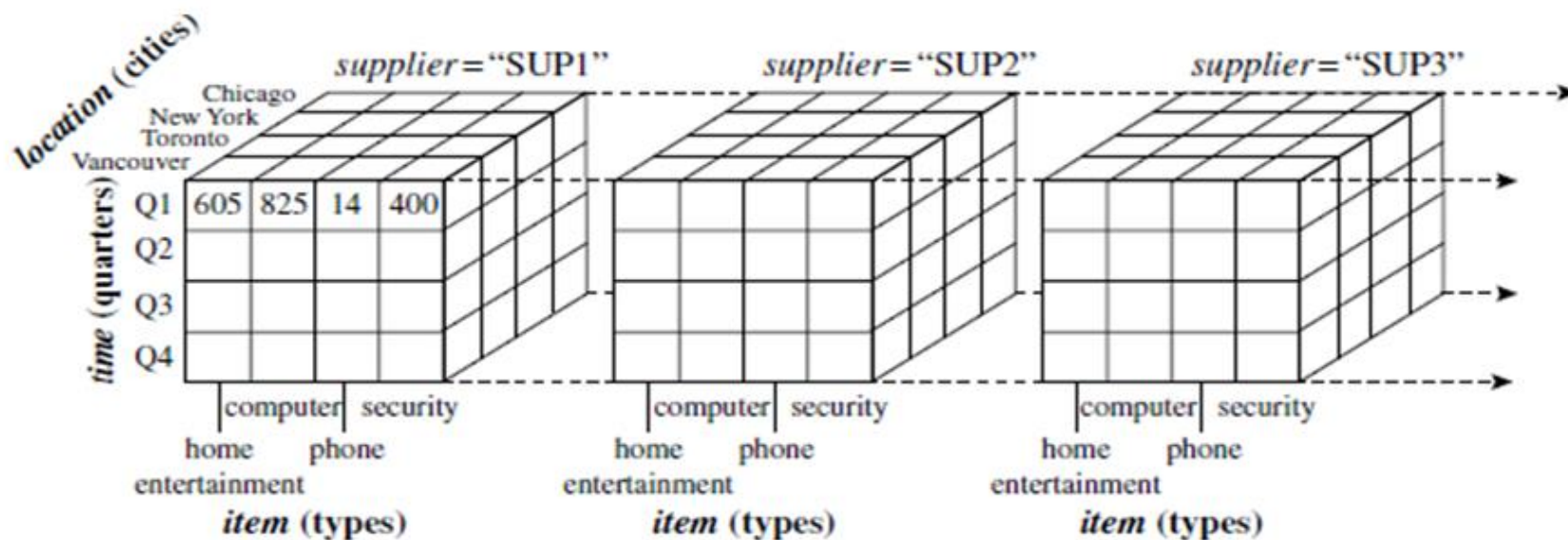
<i>location</i> = "Chicago"					<i>location</i> = "New York"					<i>location</i> = "Toronto"					<i>location</i> = "Vancouver"				
<i>Item</i>					<i>Item</i>					<i>Item</i>					<i>Item</i>				
<i>home</i>					<i>home</i>					<i>home</i>					<i>home</i>				
<i>time</i>	<i>ent.</i>	<i>comp.</i>	<i>phone</i>	<i>sec.</i>	<i>ent.</i>	<i>comp.</i>	<i>phone</i>	<i>sec.</i>		<i>ent.</i>	<i>comp.</i>	<i>phone</i>	<i>sec.</i>		<i>ent.</i>	<i>comp.</i>	<i>phone</i>	<i>sec.</i>	
Q1	854	882	89	623	1087	968	38	872	818	746	43	591	605	825	14	400			
Q2	943	890	64	698	1130	1024	41	925	894	769	52	682	680	952	31	512			
Q3	1032	924	59	789	1034	1048	45	1002	940	795	58	728	812	1023	30	501			
Q4	1129	992	63	870	1142	1091	54	984	978	864	59	784	927	1038	38	580			

Q1 854 882 89 623 1087 968 38 872 818 746 43 591 605 825 14 400

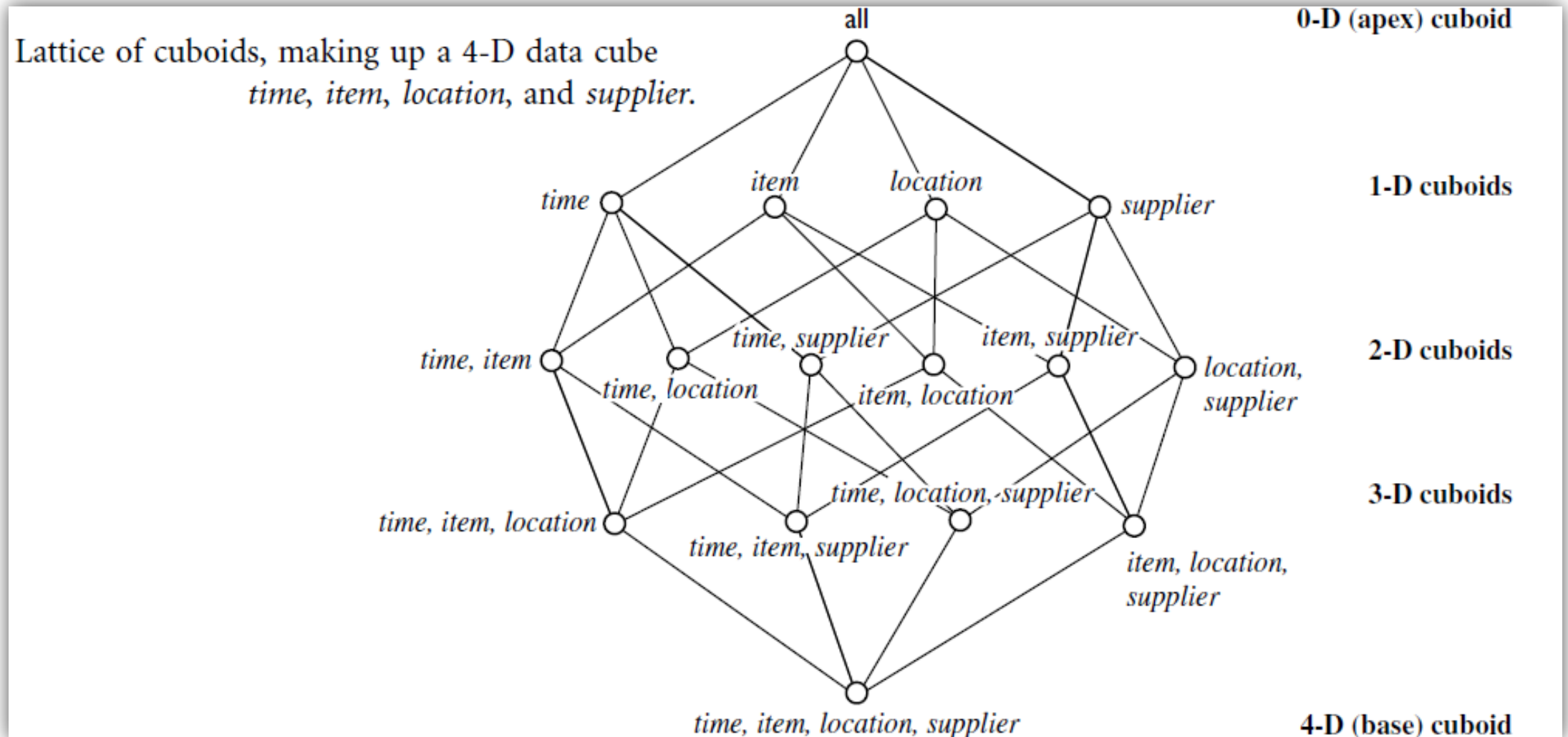


A 3-D data cube representation of the data

A 4-D data cube representation of sales data, according to *time, item, location, and supplier*.



- A **lattice of cuboids** forming a data cube for the dimensions *time*, *item*, *location*, and *supplier*.



Stars, Snowflakes, and Fact Constellations: Schemas for Multidimensional Data

- The most popular data model for a data warehouse is a **multidimensional model**

❖ **Star Schema:** The data warehouse contains

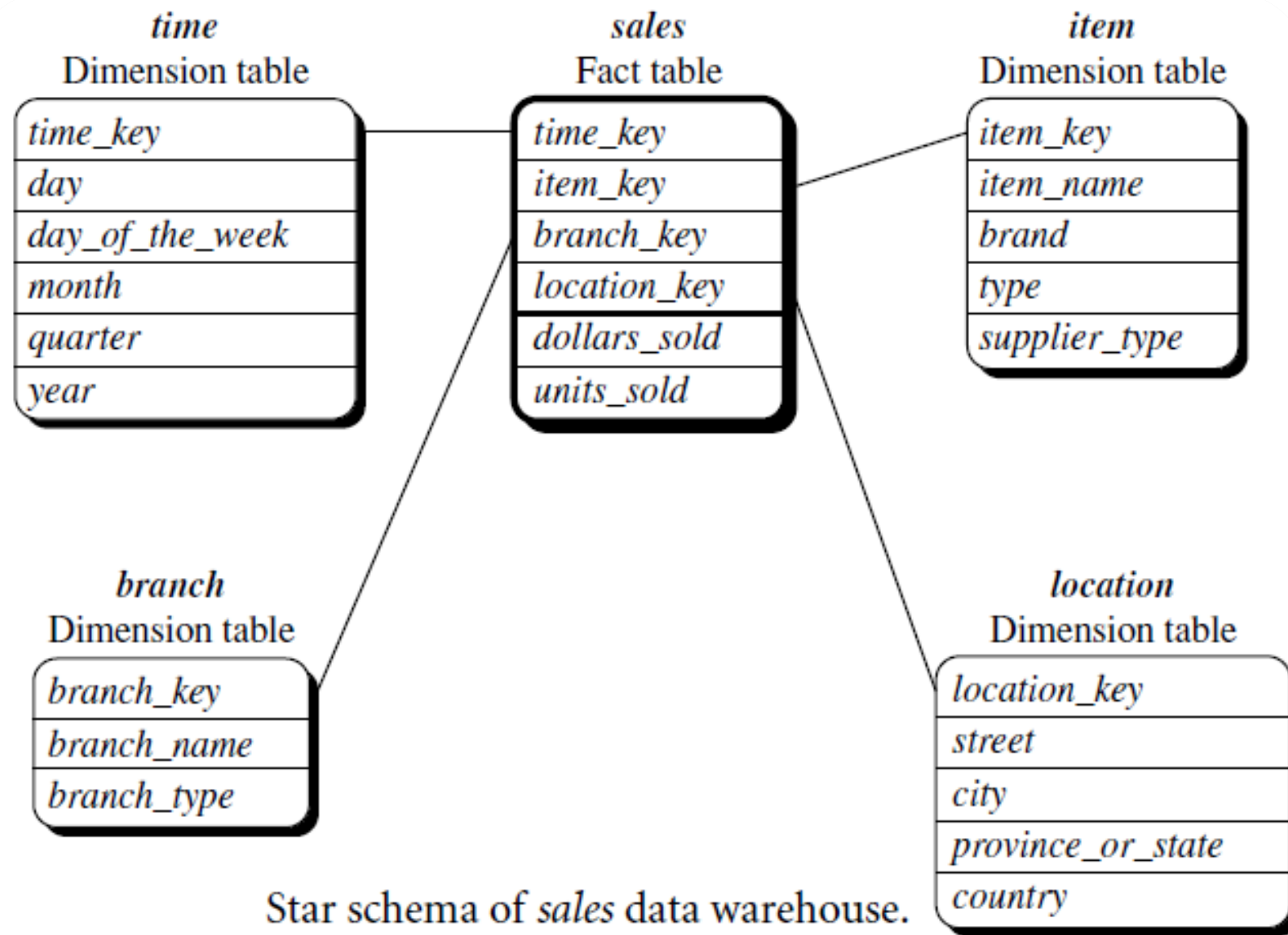
- A large central table (**fact table**) containing the bulk of the data, with no redundancy
- A set of smaller attendant tables (**dimension tables**), one for each dimension.

❖ **Snowflake schema:**

- The snowflake schema is a **variant of the star schema** model
- Some dimension tables are **normalized**, thereby further **splitting** the data into additional tables.
- The resulting schema graph forms a shape similar to a snowflake.

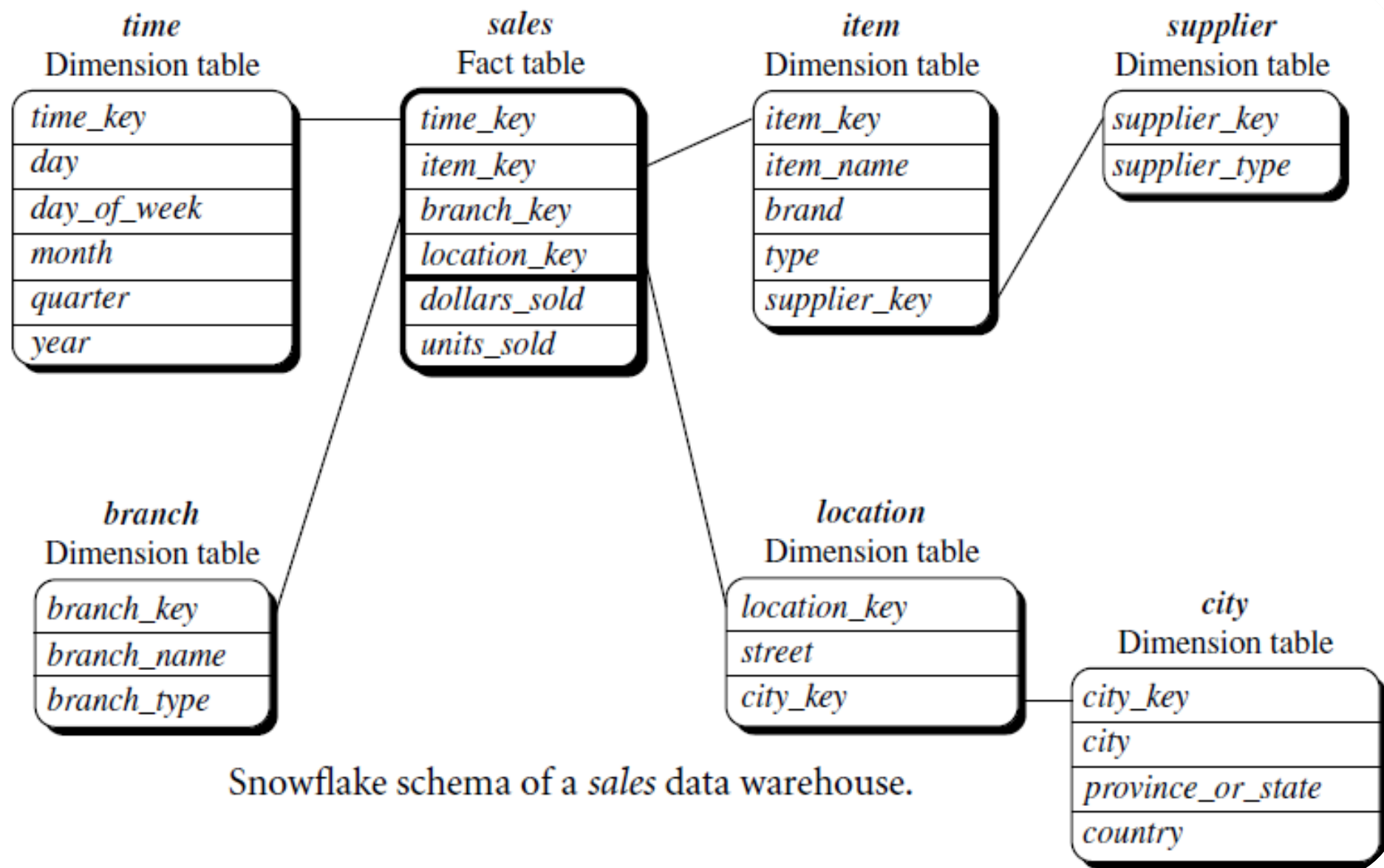
❖ **Fact constellation:**

- Sophisticated applications may require **multiple fact tables** to *share* dimension tables.
- This kind of schema can be viewed as **a collection of stars**, and hence is called a **galaxy schema** or a **fact constellation**

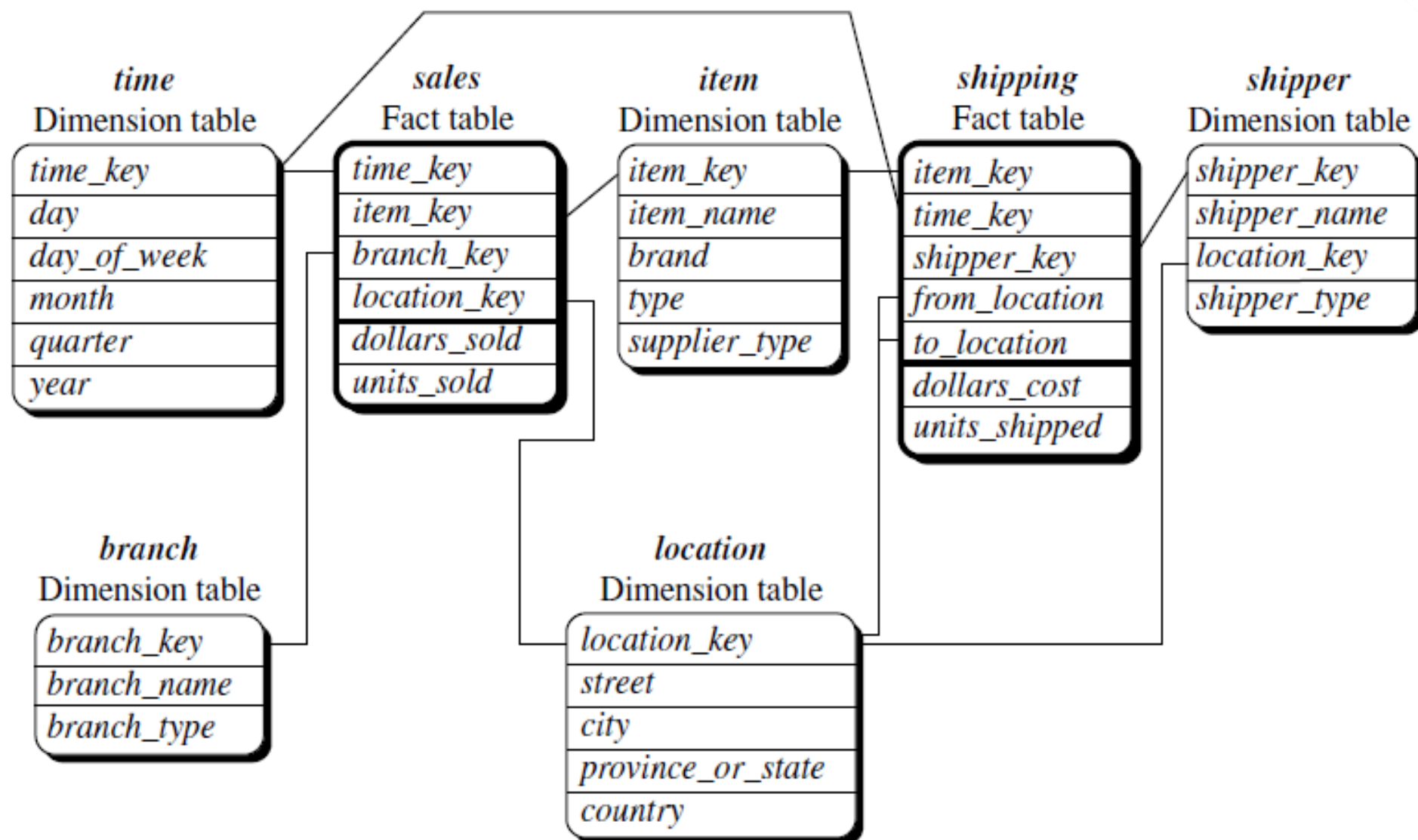


Star schema of sales data warehouse.

company
business_line



Snowflake schema of a *sales* data warehouse.

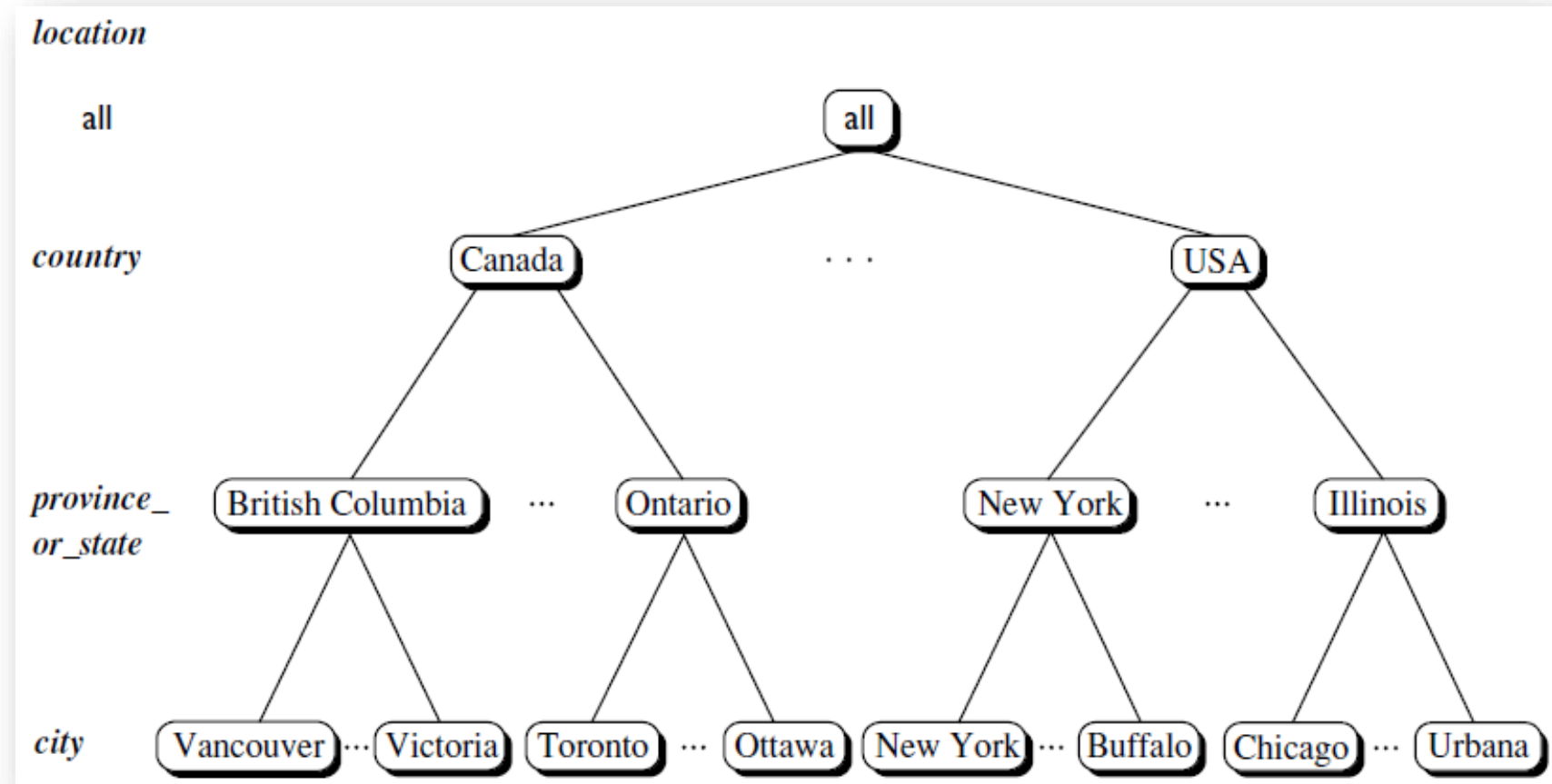


Fact constellation schema of a sales and shipping data warehouse.

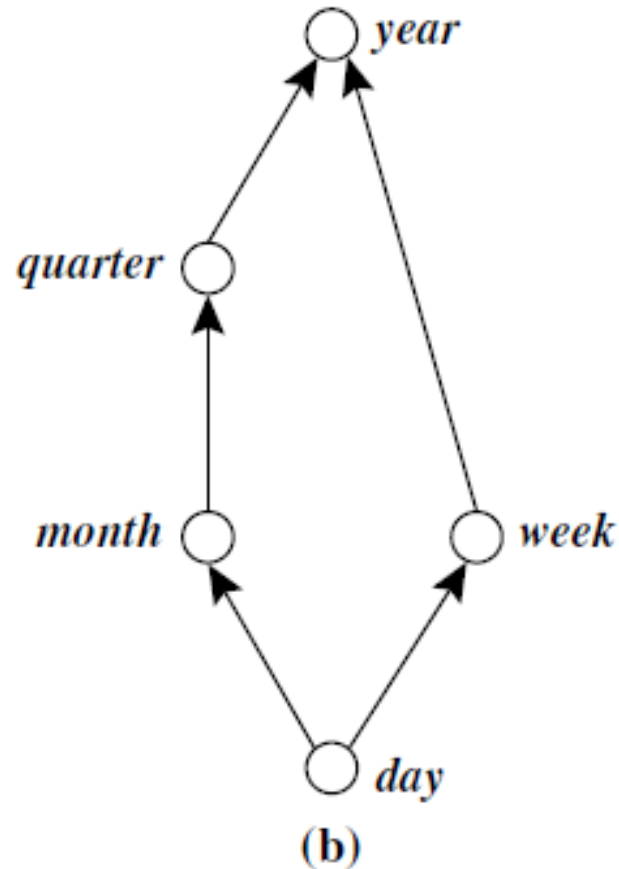
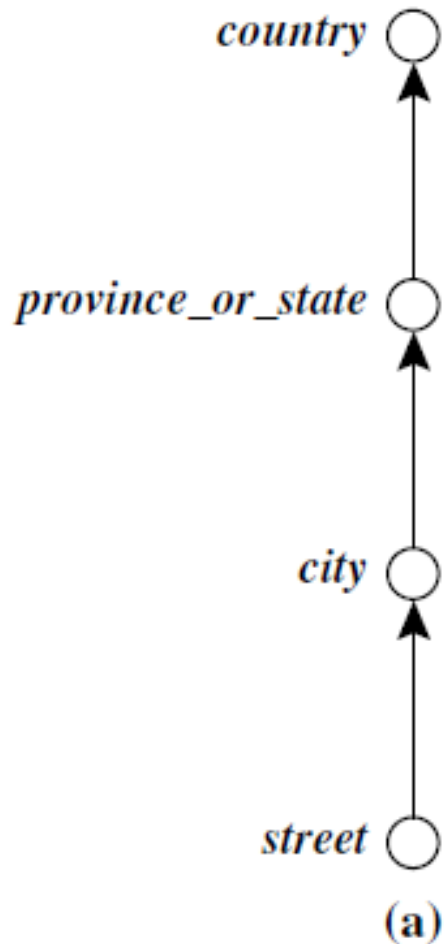
Fact constellation schema of a sales and shipping data warehouse.

Dimensions: The Role of Concept Hierarchies

- A **concept hierarchy** defines a **sequence of mappings** from a set of low-level concepts to higher-level, more general concepts.
- Many concept hierarchies are **implicit** within the database schema.
- Dimension :
 - **location**
- Attributes
 - **number**
 - **street**
 - **city**
 - **province or state**
 - **zip code**
 - **country**



- The attributes of a dimension may be organized in **a partial order**, forming a **lattice**.
- E.g., “*day* < {*month* < *quarter*; *week*} < *year*”



Hierarchical and lattice structures of attributes in warehouse dimensions:

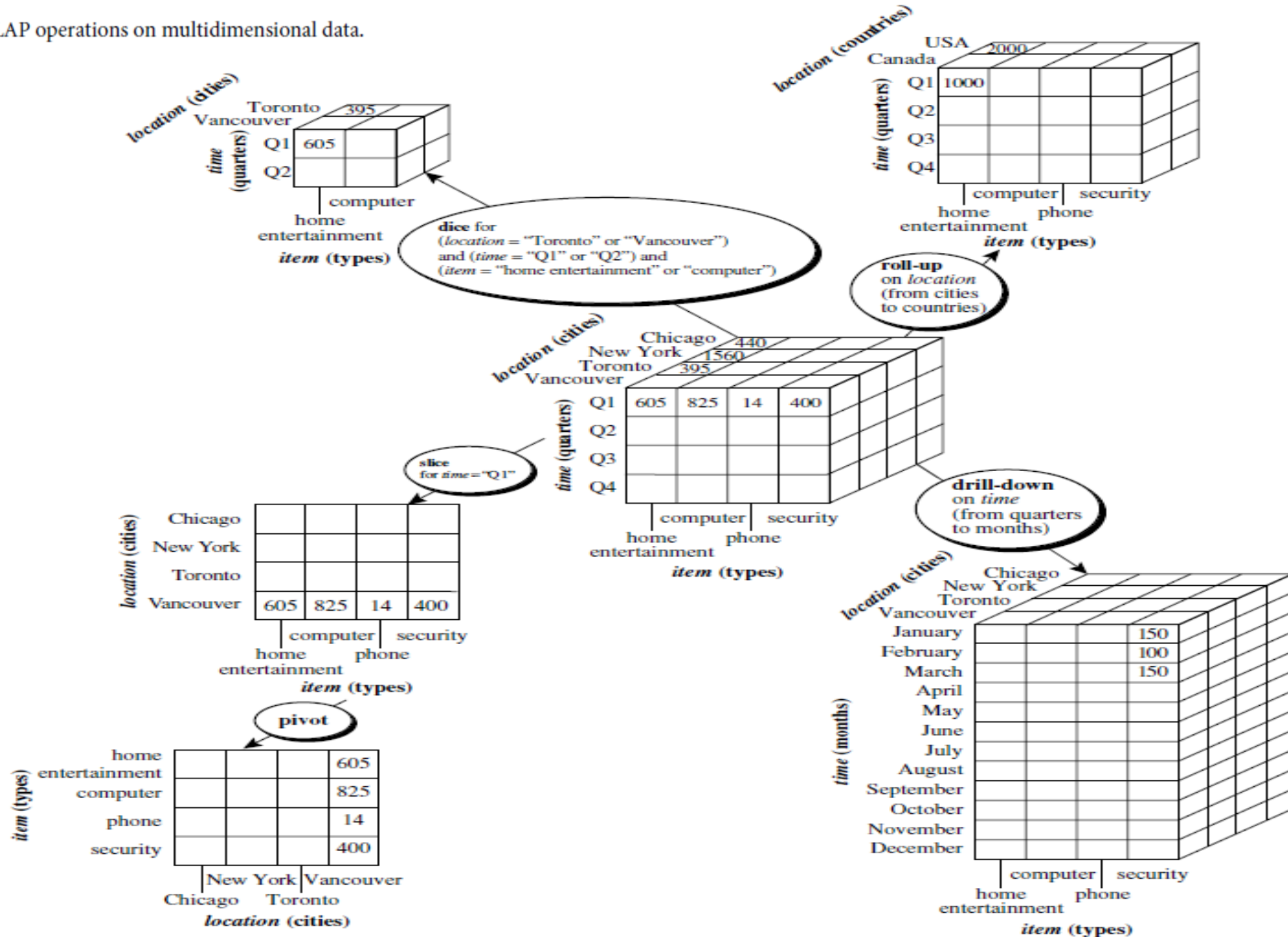
(a) a hierarchy for *location*

(b) a lattice for *time*.

Typical OLAP Operations

- In the **multidimensional model**, **data** are organized into **multiple dimensions**, and each dimension contains **multiple levels of abstraction** defined by concept hierarchies.
 - A **number of OLAP data cube operations** exist to materialize these different views.
1. **Roll-up:** The roll-up operation (also called the *drill-up* operation by some vendors) performs **aggregation** on a data cube, either by *climbing up a concept hierarchy* for a dimension or by *dimension reduction*.
 2. **Drill-down:** It is the **reverse** of roll-up. It **navigates** from less detailed data to more detailed data. Drill-down can be realized by either *stepping down a concept hierarchy* for a dimension or *introducing additional dimensions*.

Examples of typical OLAP operations on multidimensional data.



3) Slice and dice: The **slice** operation performs a selection on one dimension of the given cube, resulting in The **dice** operation defines a sub cube by performing a selection on two or more dimensions.

4) Pivot (rotate): *Pivot* (also called *rotate*) is a visualization operation that rotates the data axes in view to provide an alternative data presentation.

The other operations are:

5) Drill-across executes queries involving (i.e., across) more than one fact table.

6) Drill-through operation uses relational SQL facilities to drill through the bottom level of a data cube down to its back-end relational tables.

UNIT-1
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